

BFS Traversal

Step 1: Start at E

Legend

Complete

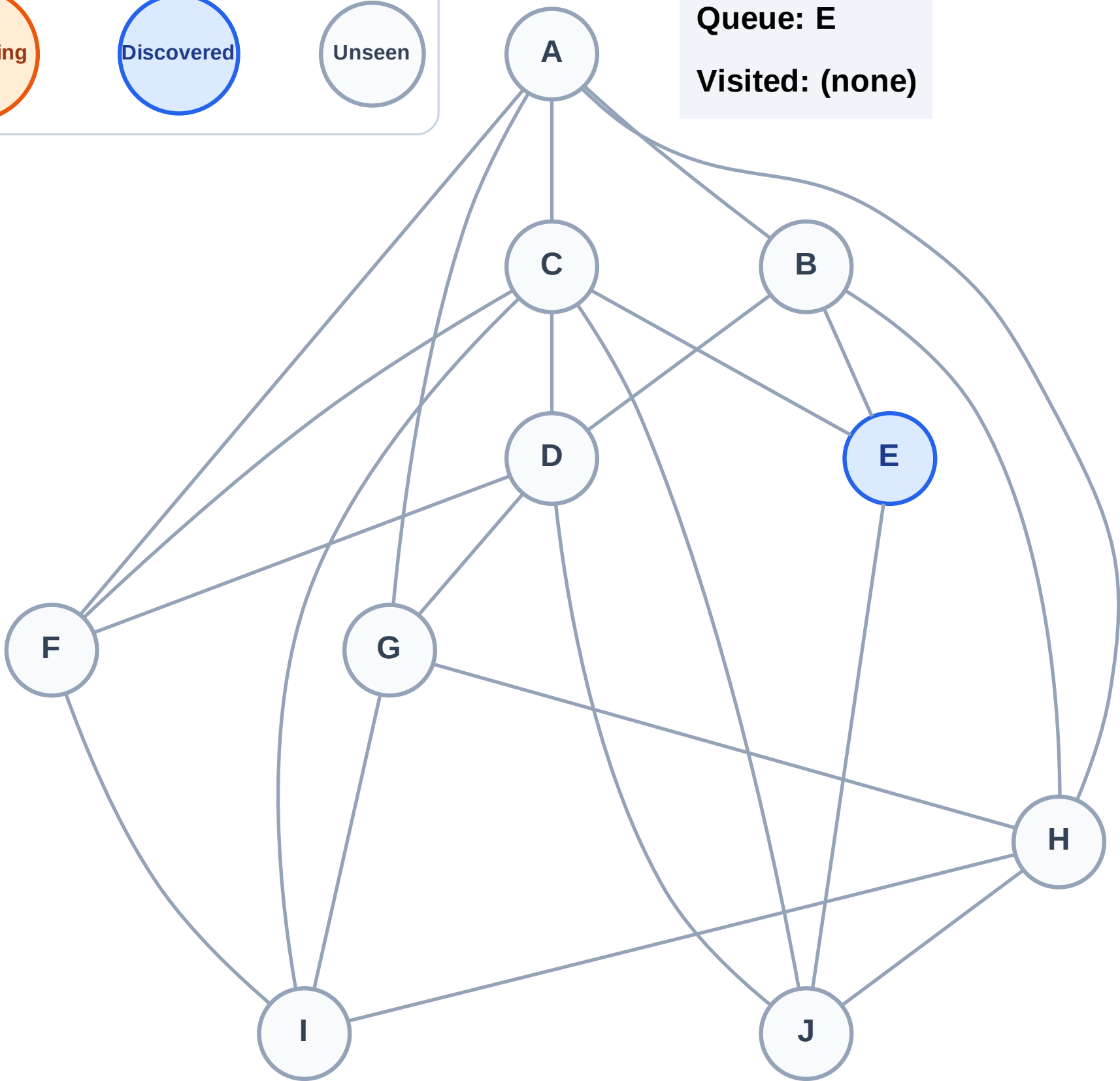
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

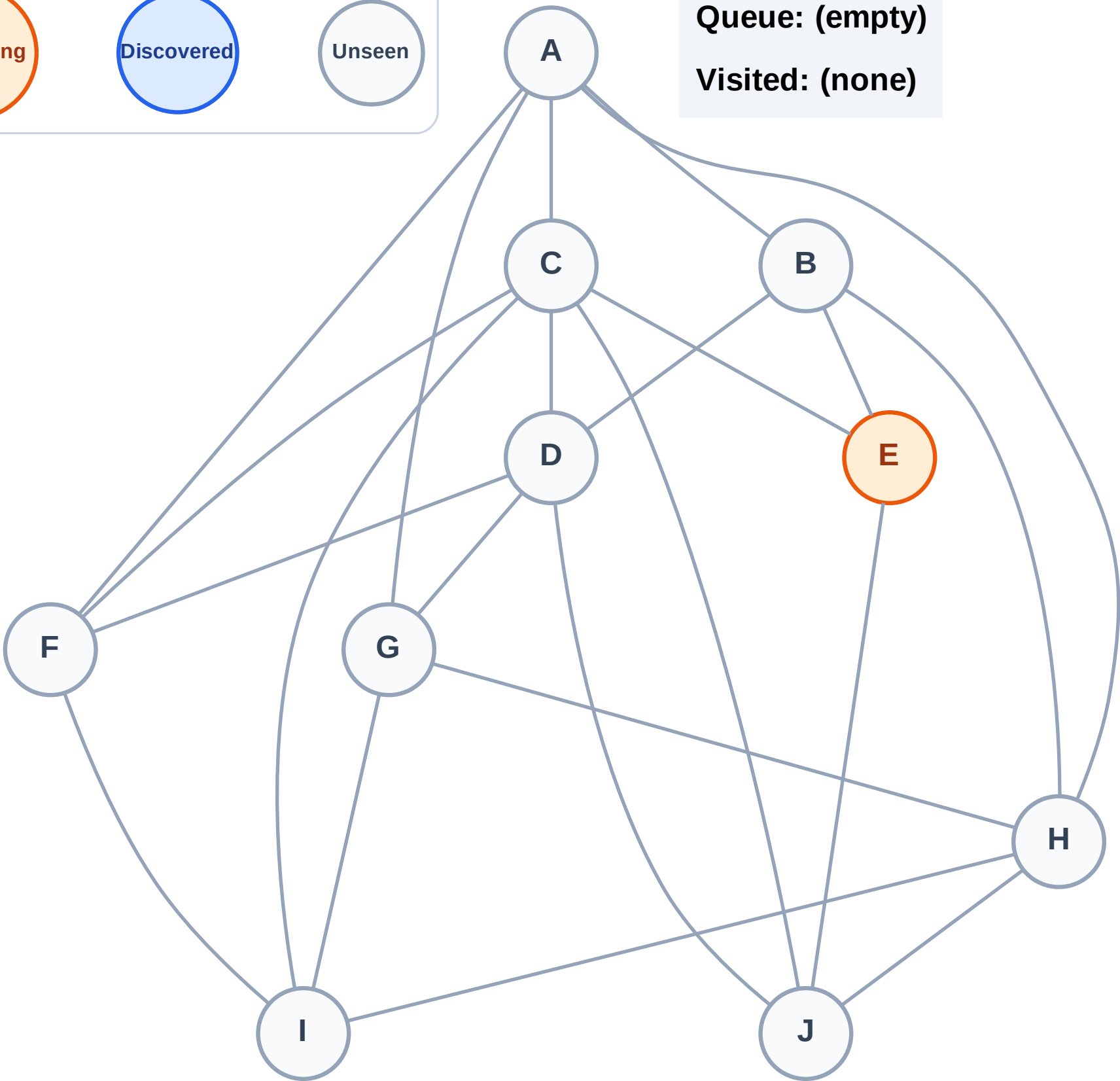
Complete

Processing

Discovered

Unseen

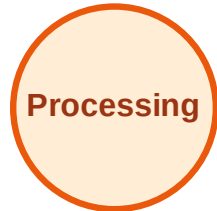
Queue: (empty)
Visited: (none)



BFS Traversal

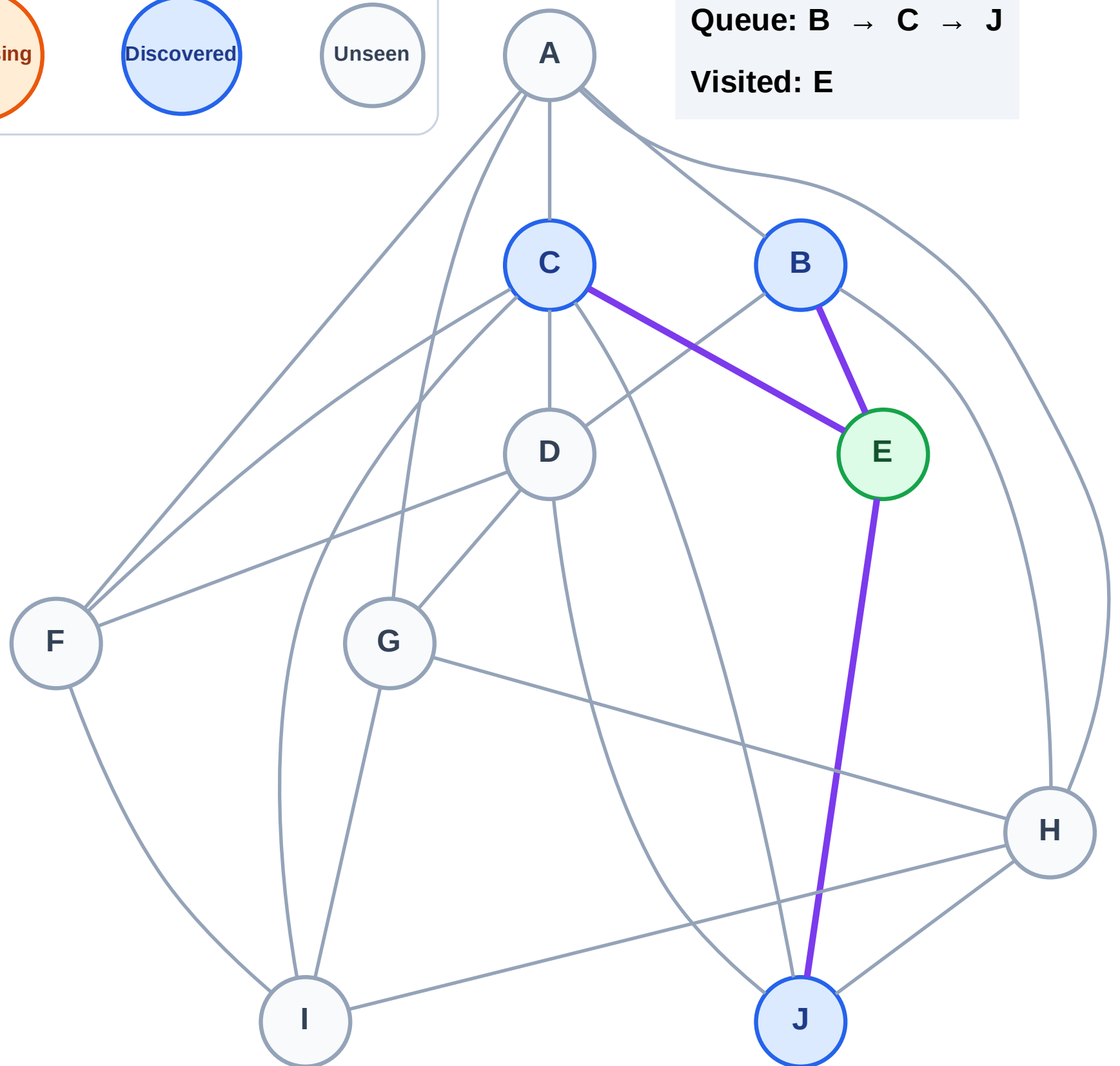
Step 3: Discover B, C, J from E

Legend



Queue: B → C → J

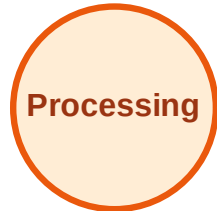
Visited: E



BFS Traversal

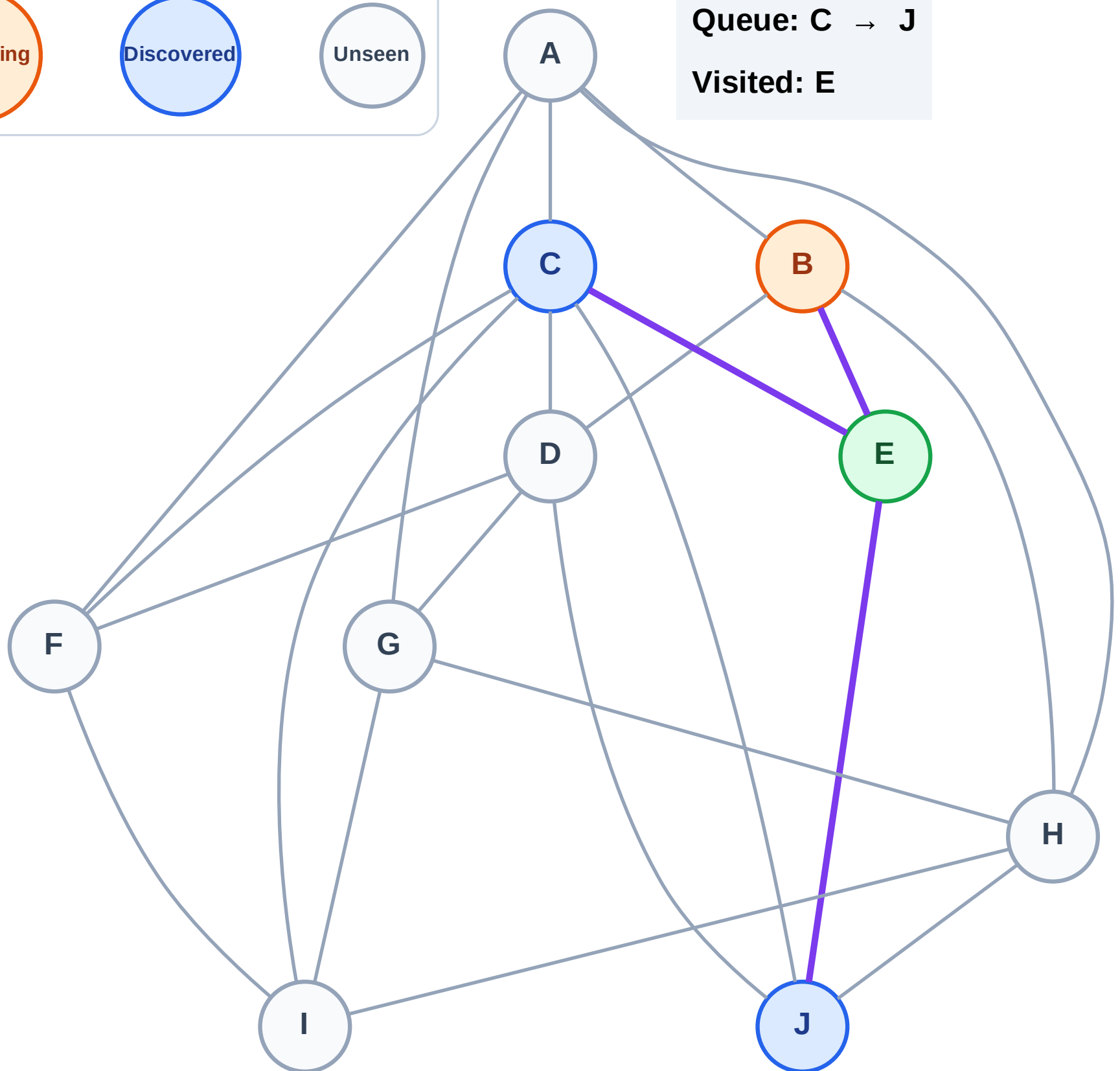
Step 4: Process node B

Legend



Queue: C → J

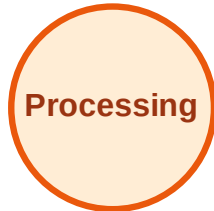
Visited: E



BFS Traversal

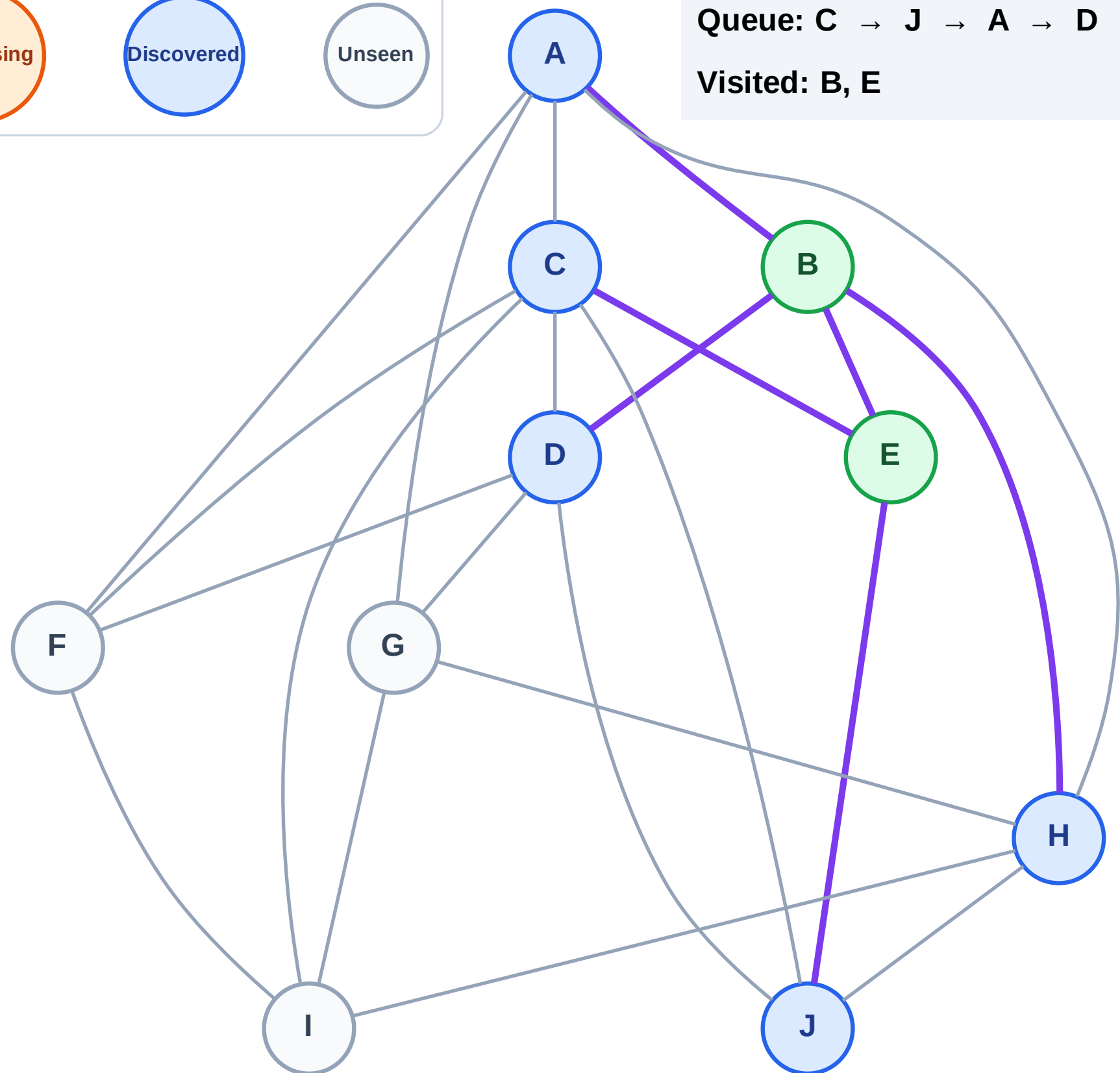
Step 5: Discover A, D, H from B

Legend



Queue: C → J → A → D → H

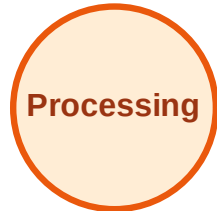
Visited: B, E



BFS Traversal

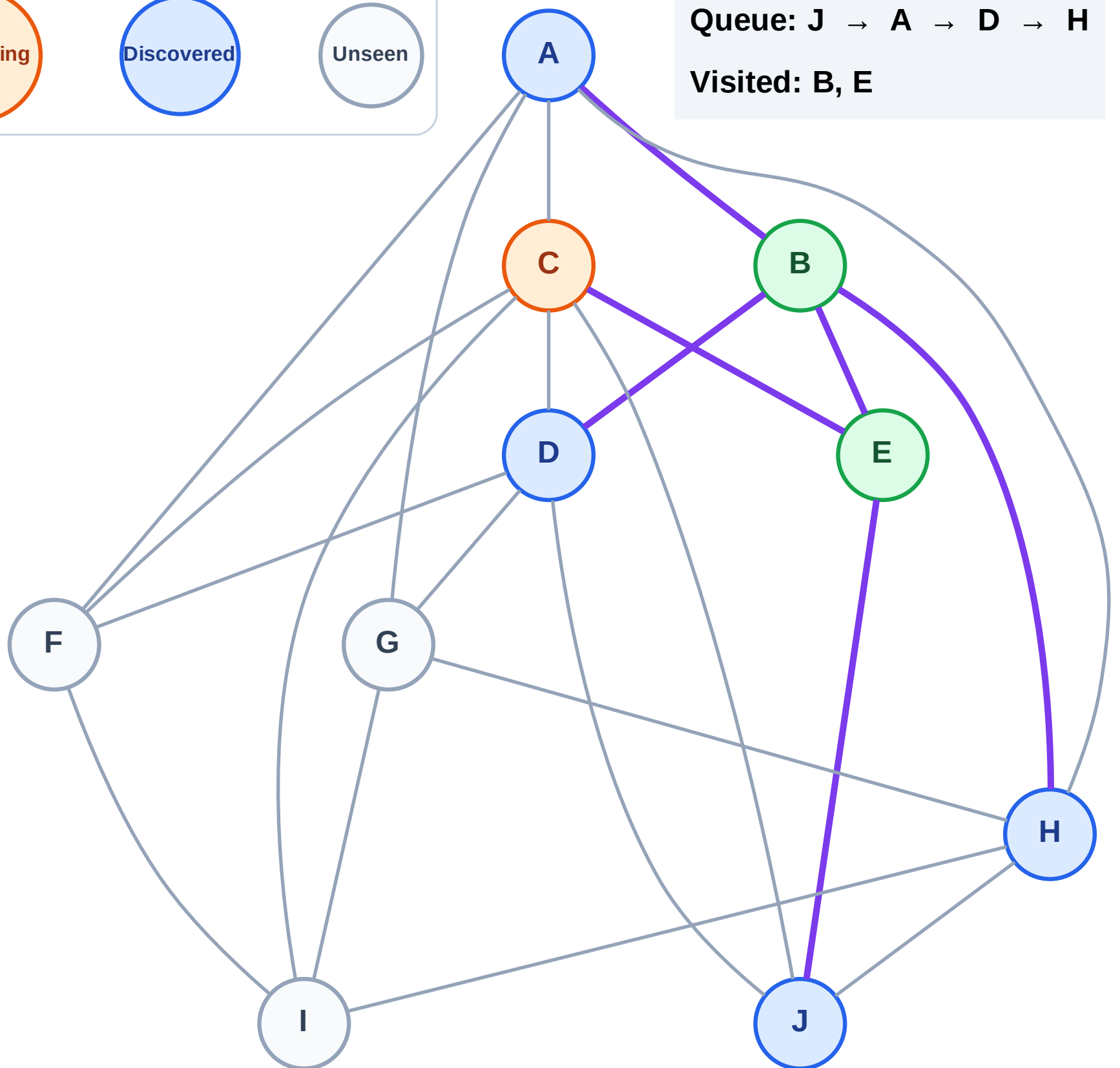
Step 6: Process node C

Legend



Queue: J → A → D → H

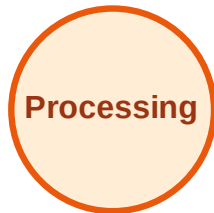
Visited: B, E



BFS Traversal

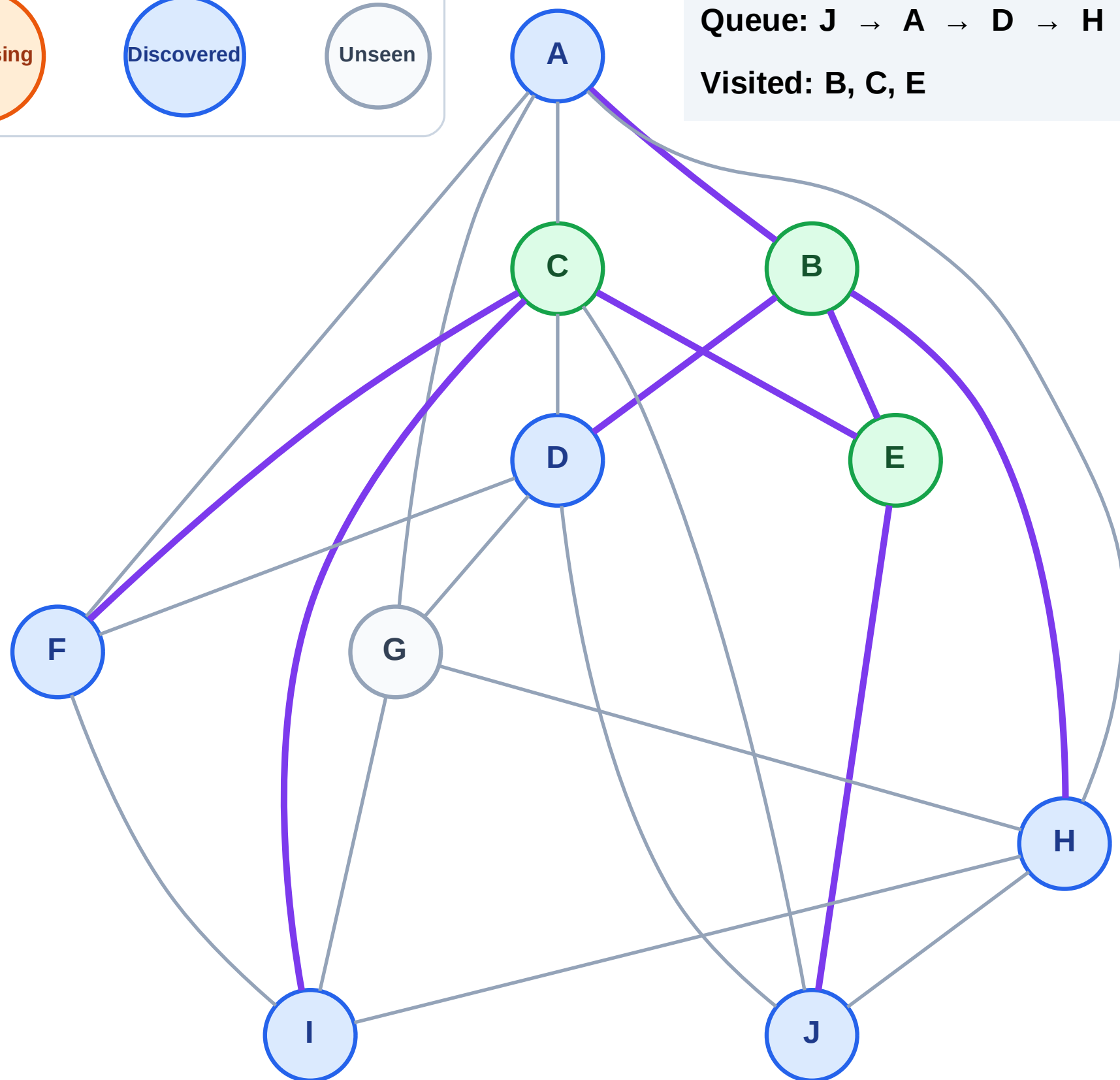
Step 7: Discover F, I from C

Legend



Queue: J → A → D → H → F → I

Visited: B, C, E



BFS Traversal

Step 8: Process node J

Legend

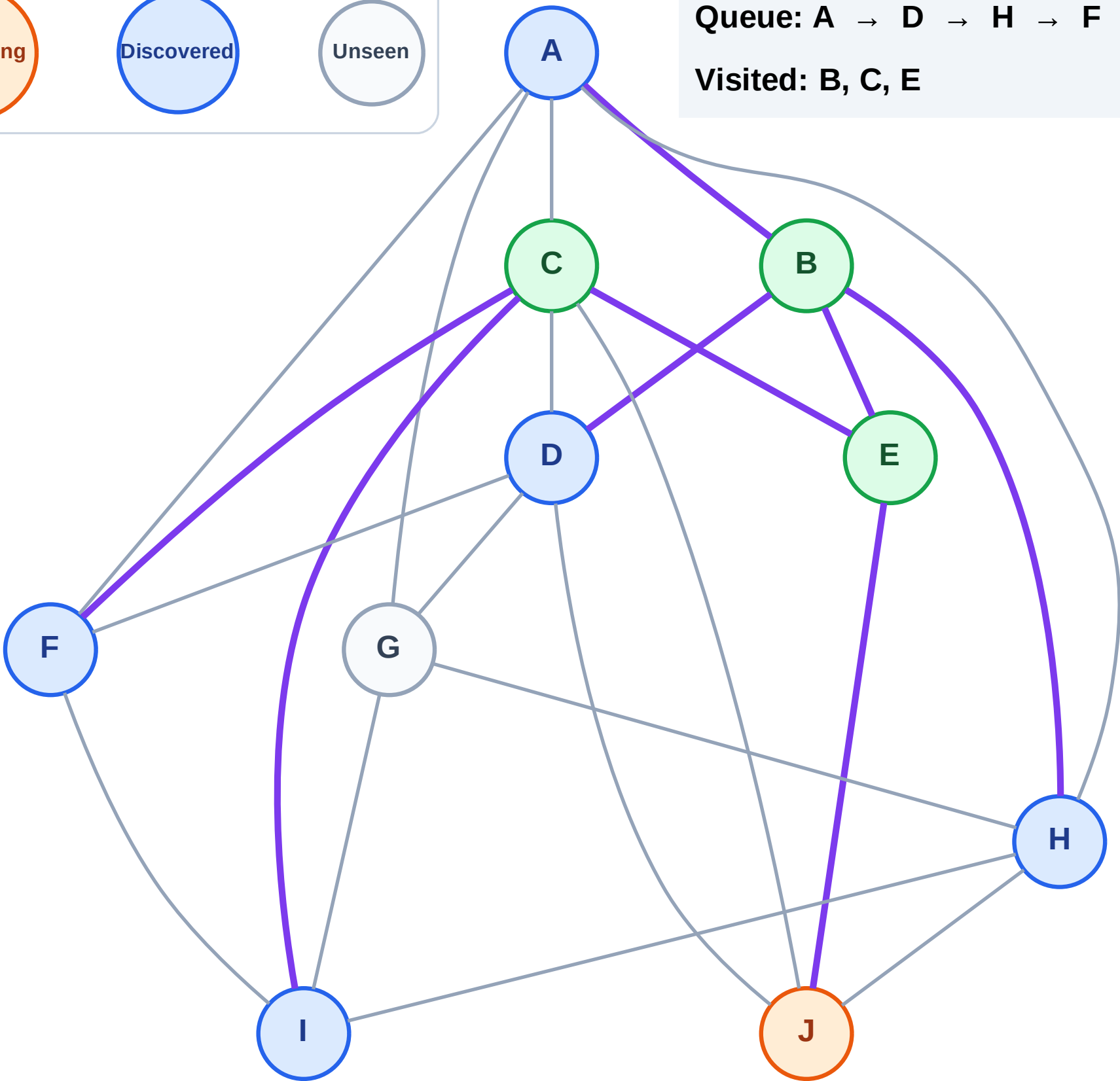
Complete

Processing

Discovered

Unseen

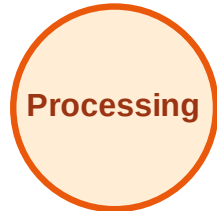
Queue: A → D → H → F → I
Visited: B, C, E



BFS Traversal

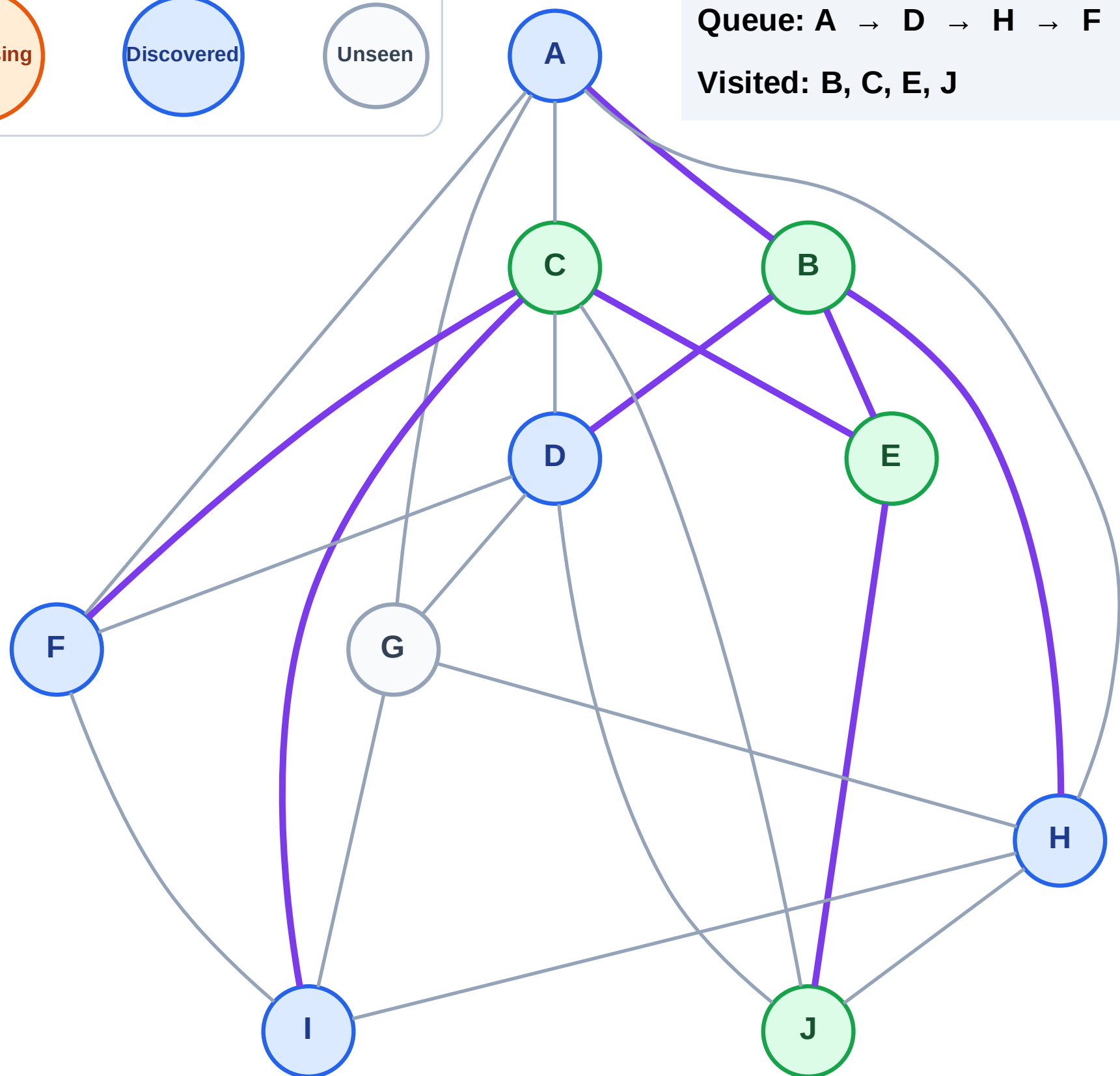
Step 9: No new neighbors from J

Legend



Queue: A → D → H → F → I

Visited: B, C, E, J



BFS Traversal

Step 10: Process node A

Legend

Complete

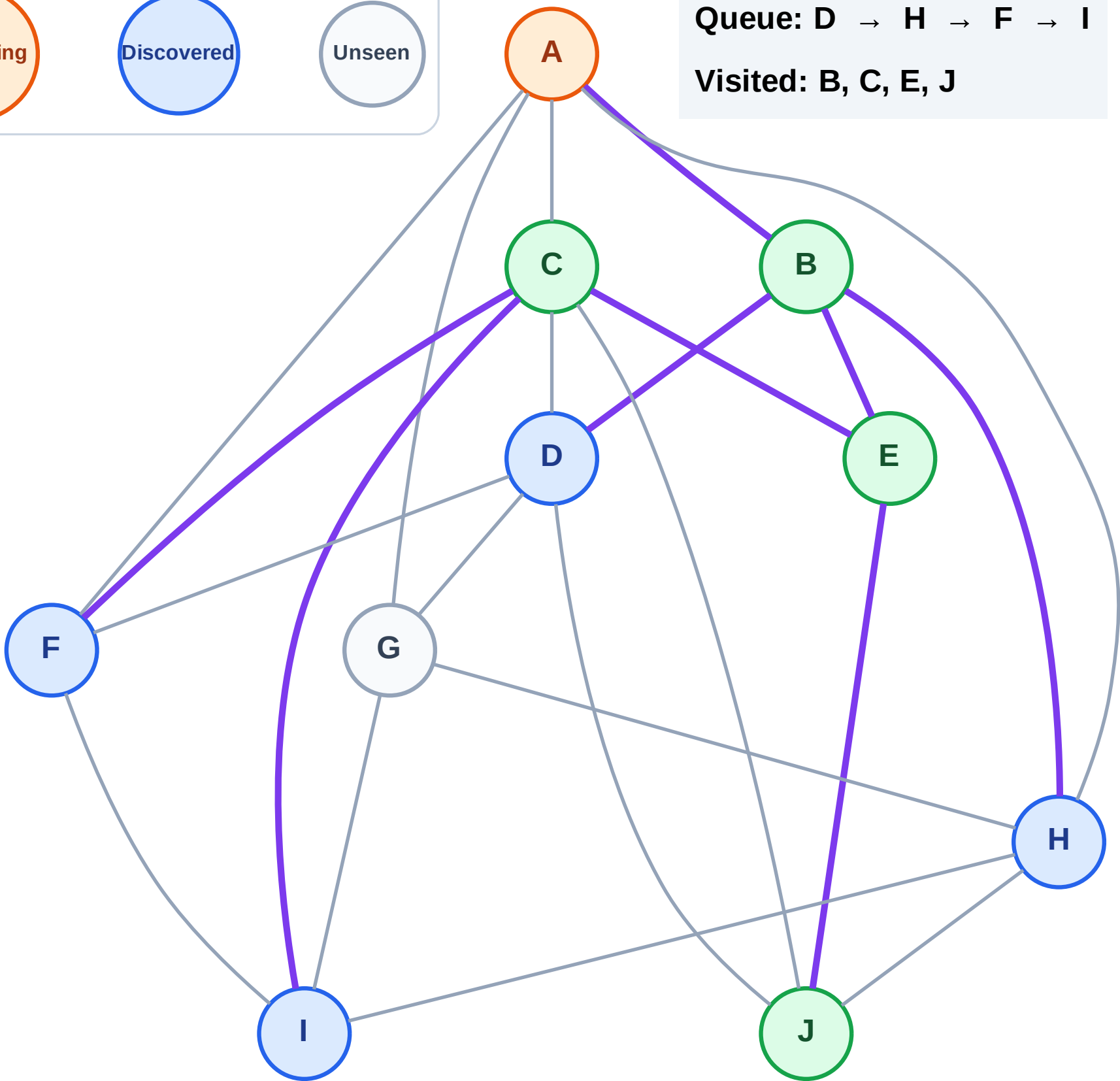
Processing

Discovered

Unseen

Queue: D → H → F → I

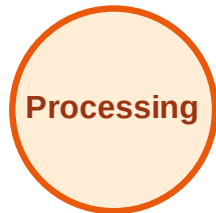
Visited: B, C, E, J



BFS Traversal

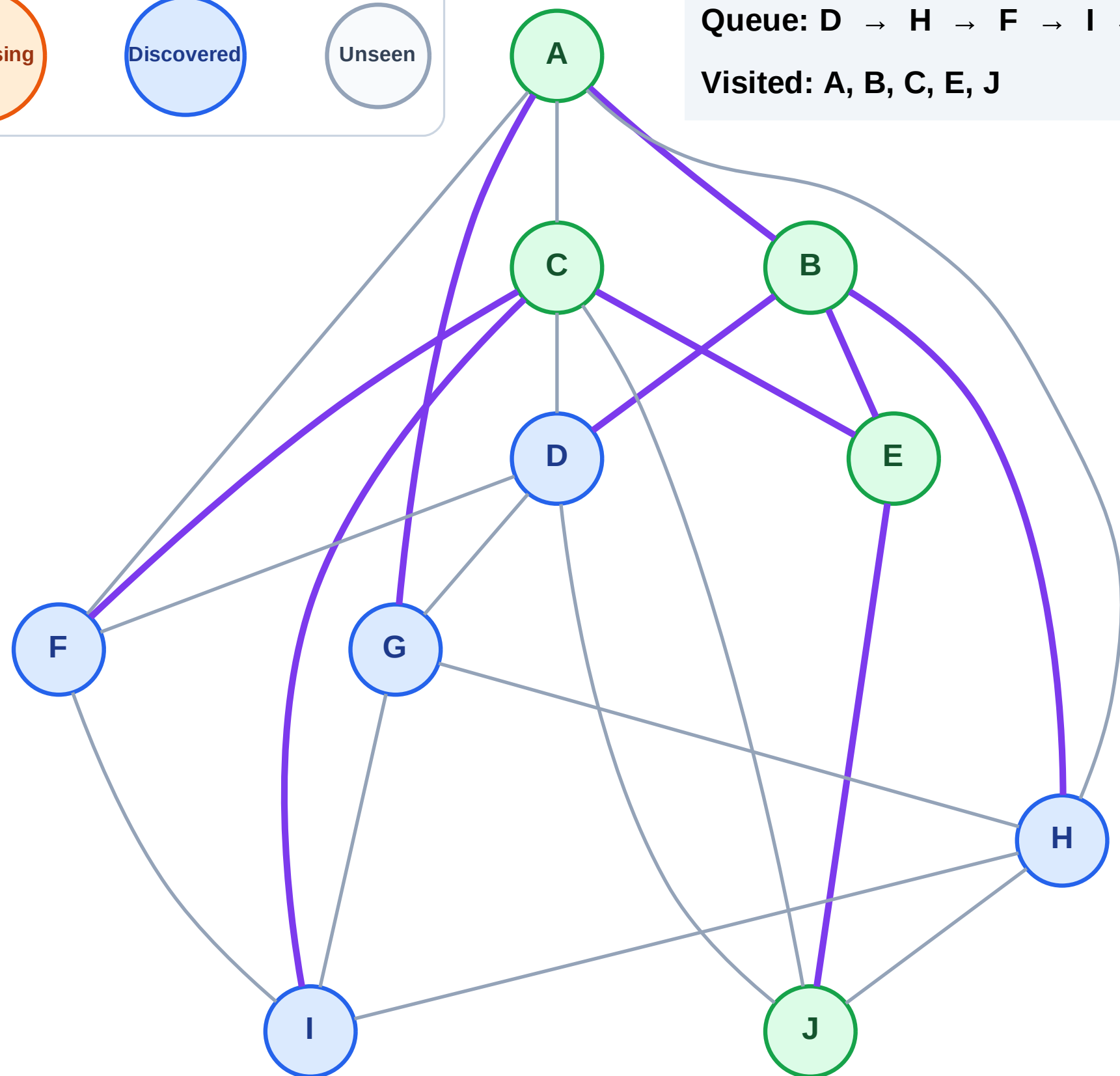
Step 11: Discover G from A

Legend



Queue: D → H → F → I → G

Visited: A, B, C, E, J



BFS Traversal

Step 12: Process node D

Legend

Complete

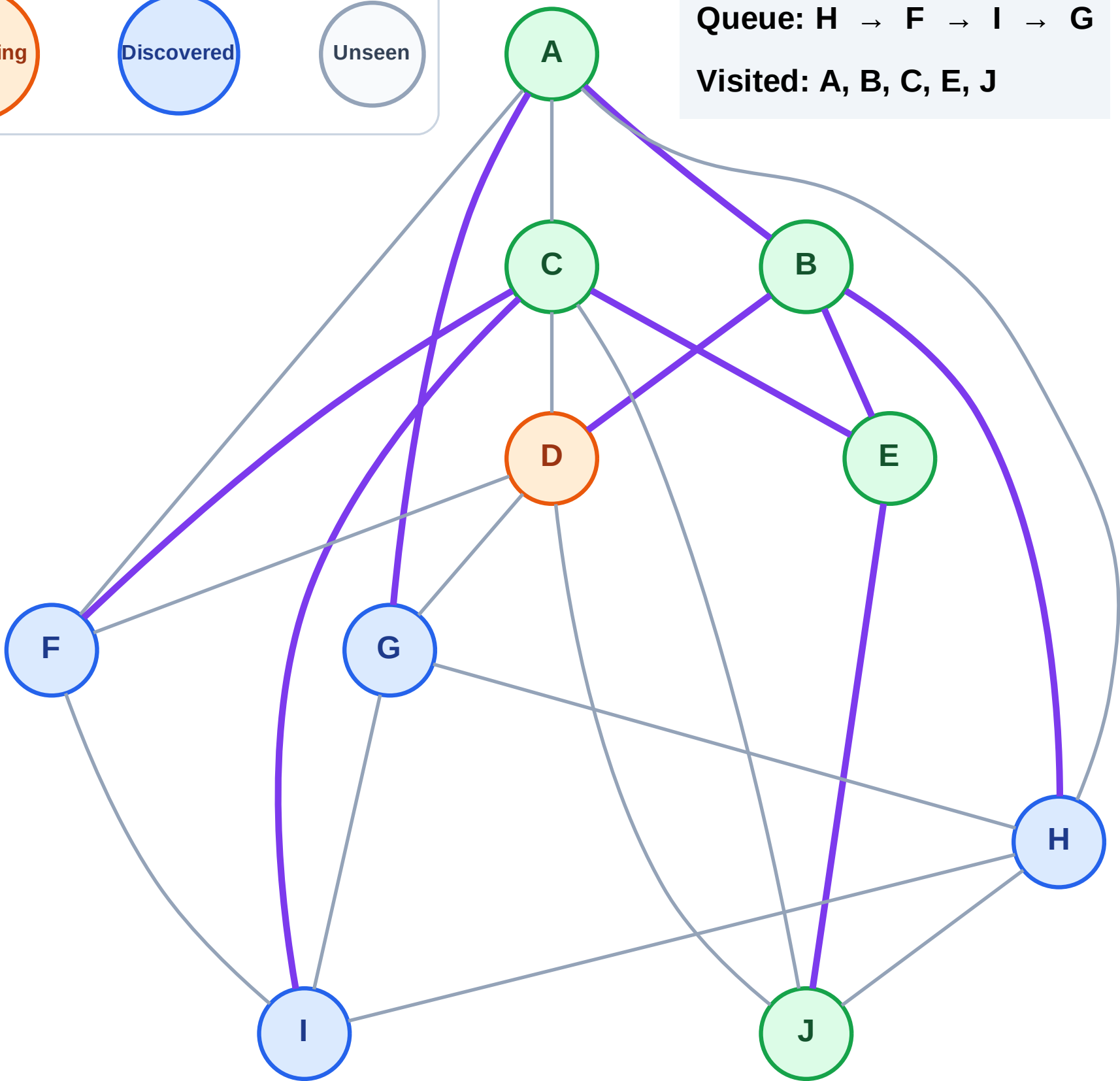
Processing

Discovered

Unseen

Queue: H → F → I → G

Visited: A, B, C, E, J

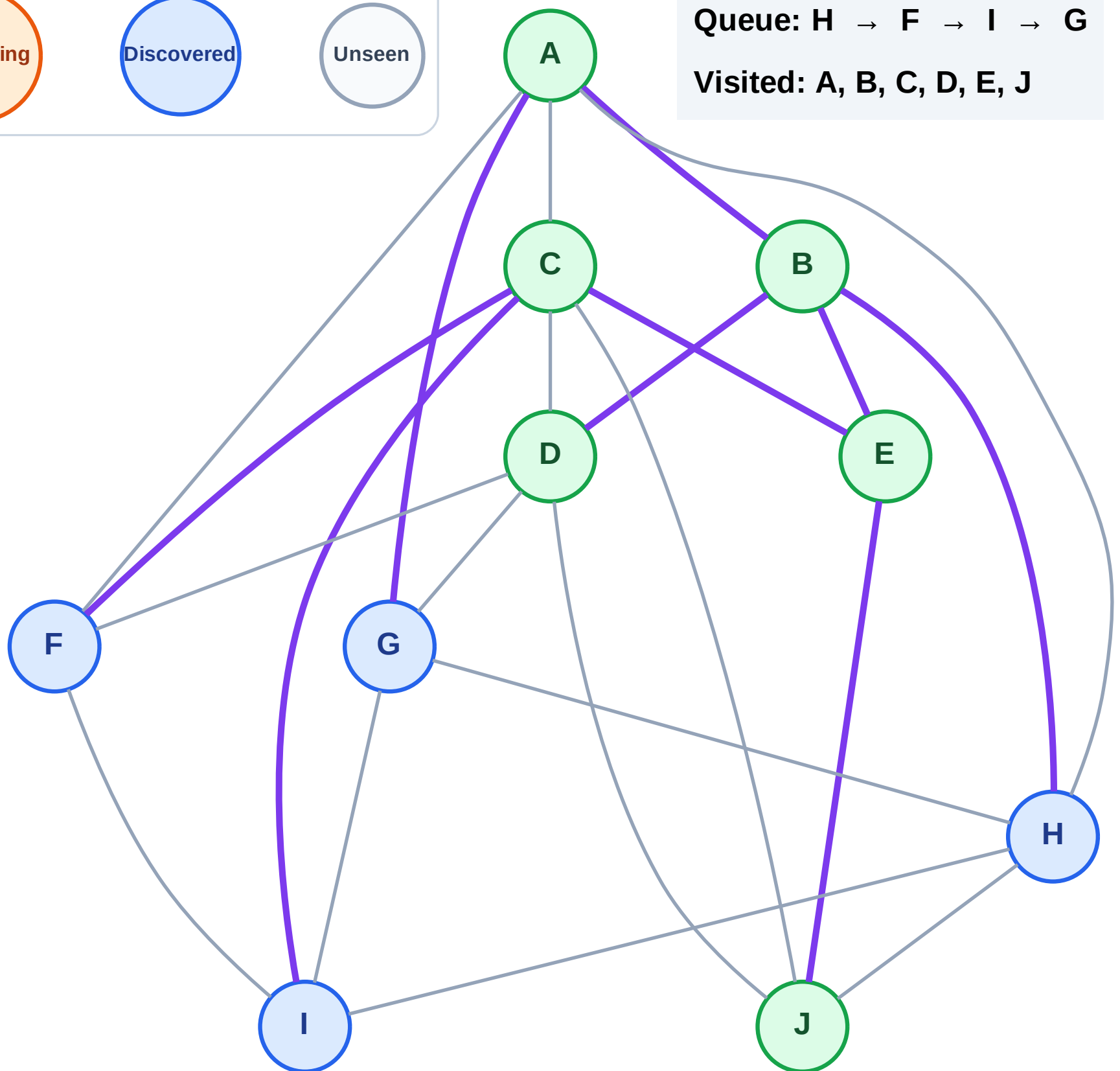


Step 13: No new neighbors from D

Legend



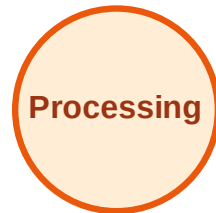
Visited: A, B, C, D, E, J



BFS Traversal

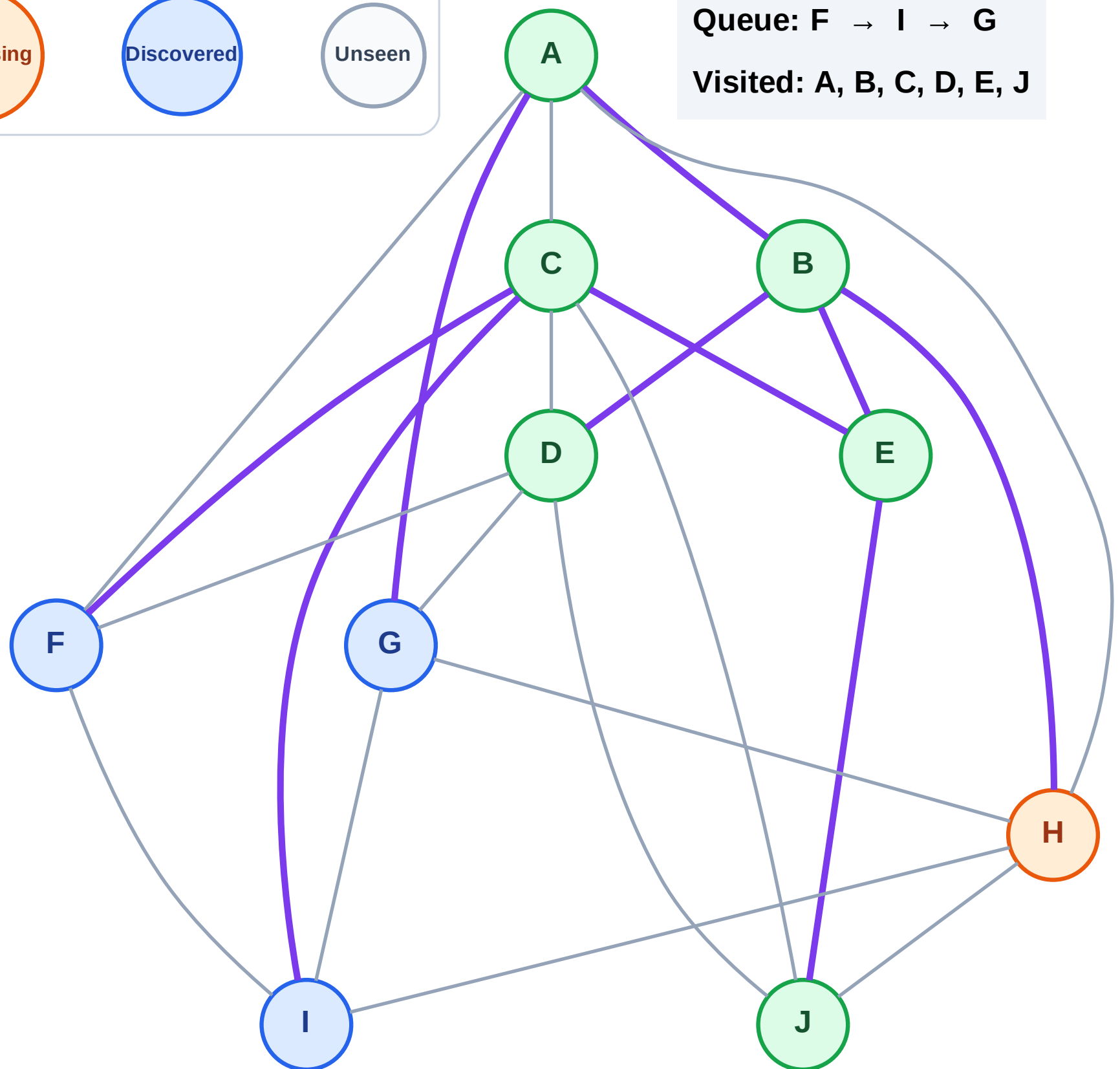
Step 14: Process node H

Legend



Queue: F → I → G

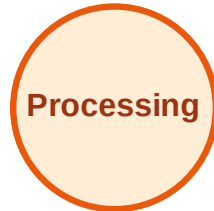
Visited: A, B, C, D, E, J



BFS Traversal

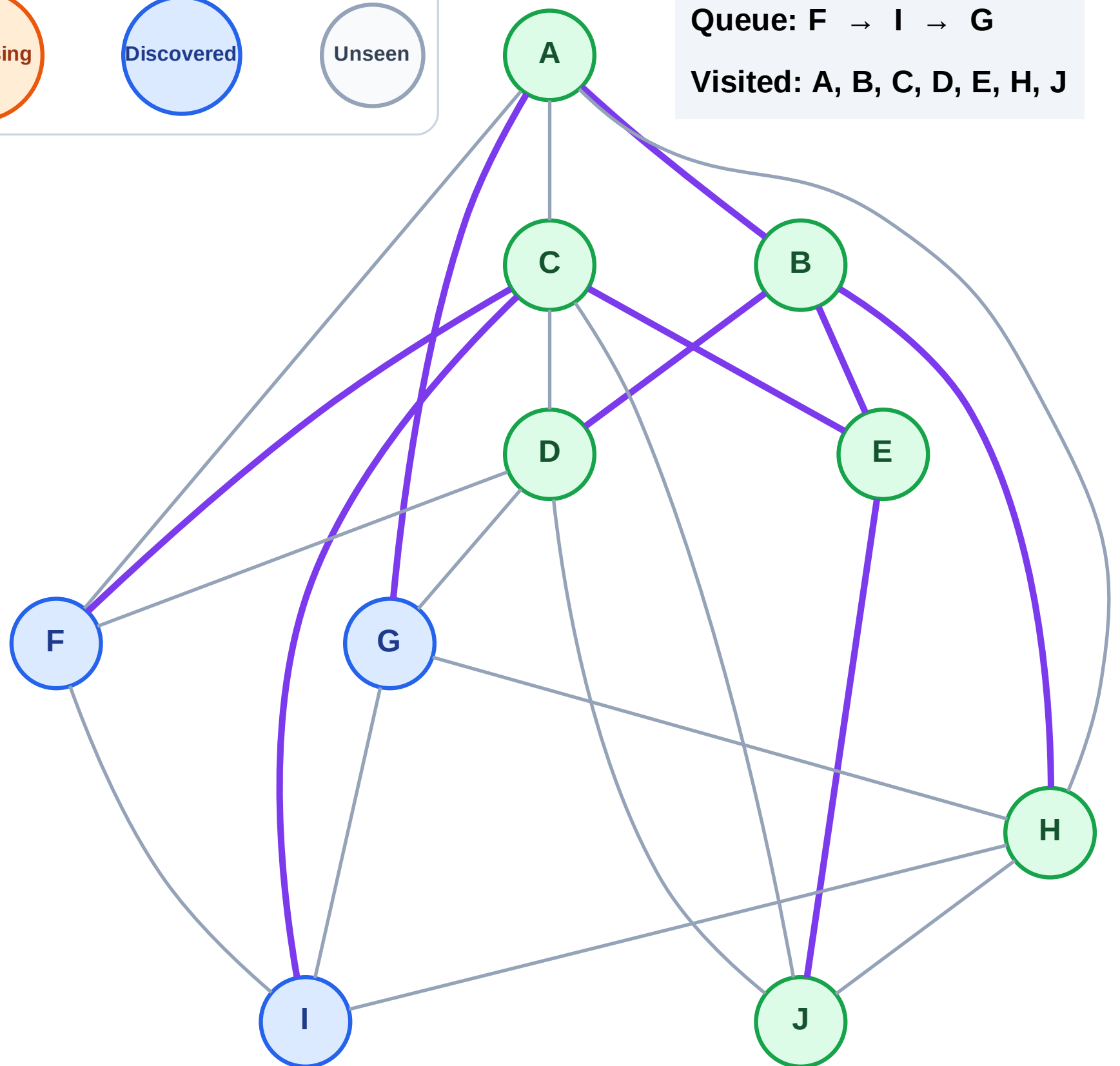
Step 15: No new neighbors from H

Legend



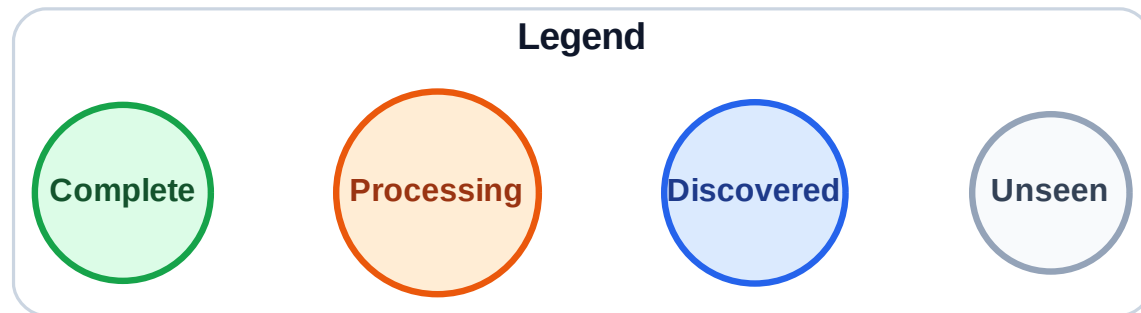
Queue: F → I → G

Visited: A, B, C, D, E, H, J



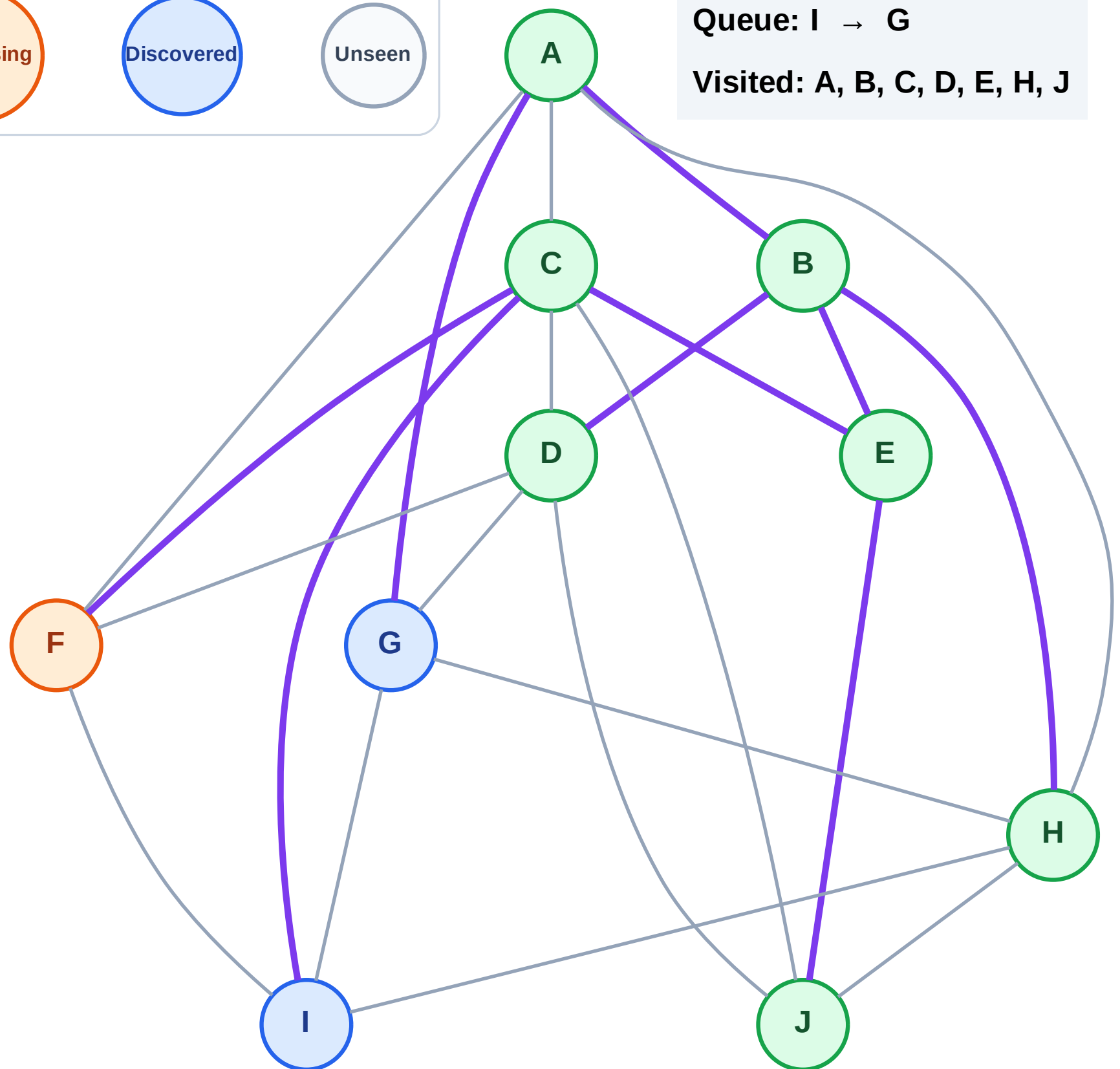
Step 16: Process node F

Step 16: Process node F



Queue: I → G

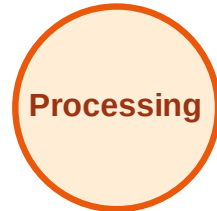
Visited: A, B, C, D, E, H, J



BFS Traversal

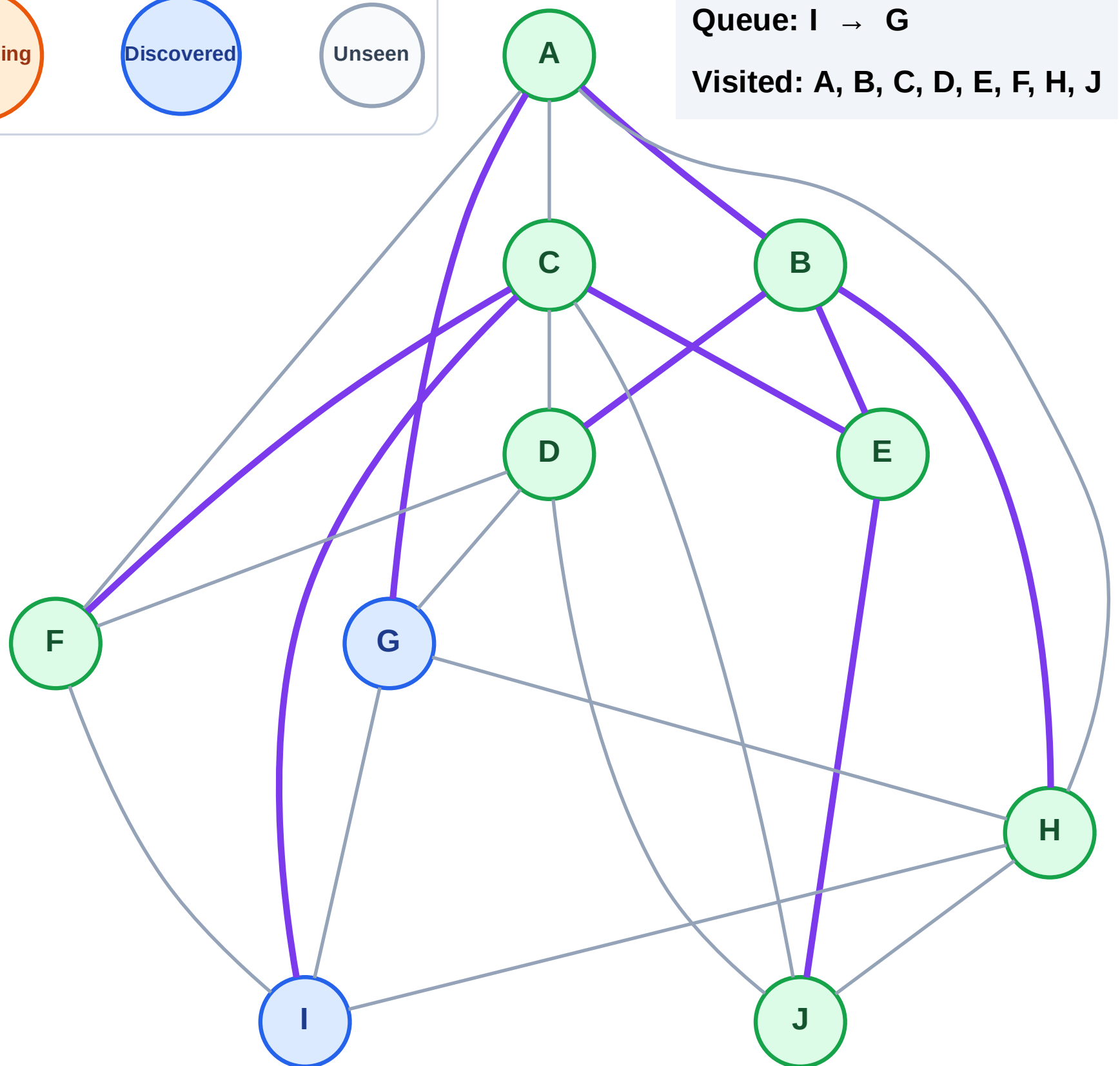
Step 17: No new neighbors from F

Legend



Queue: I → G

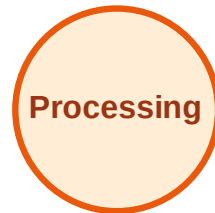
Visited: A, B, C, D, E, F, H, J



BFS Traversal

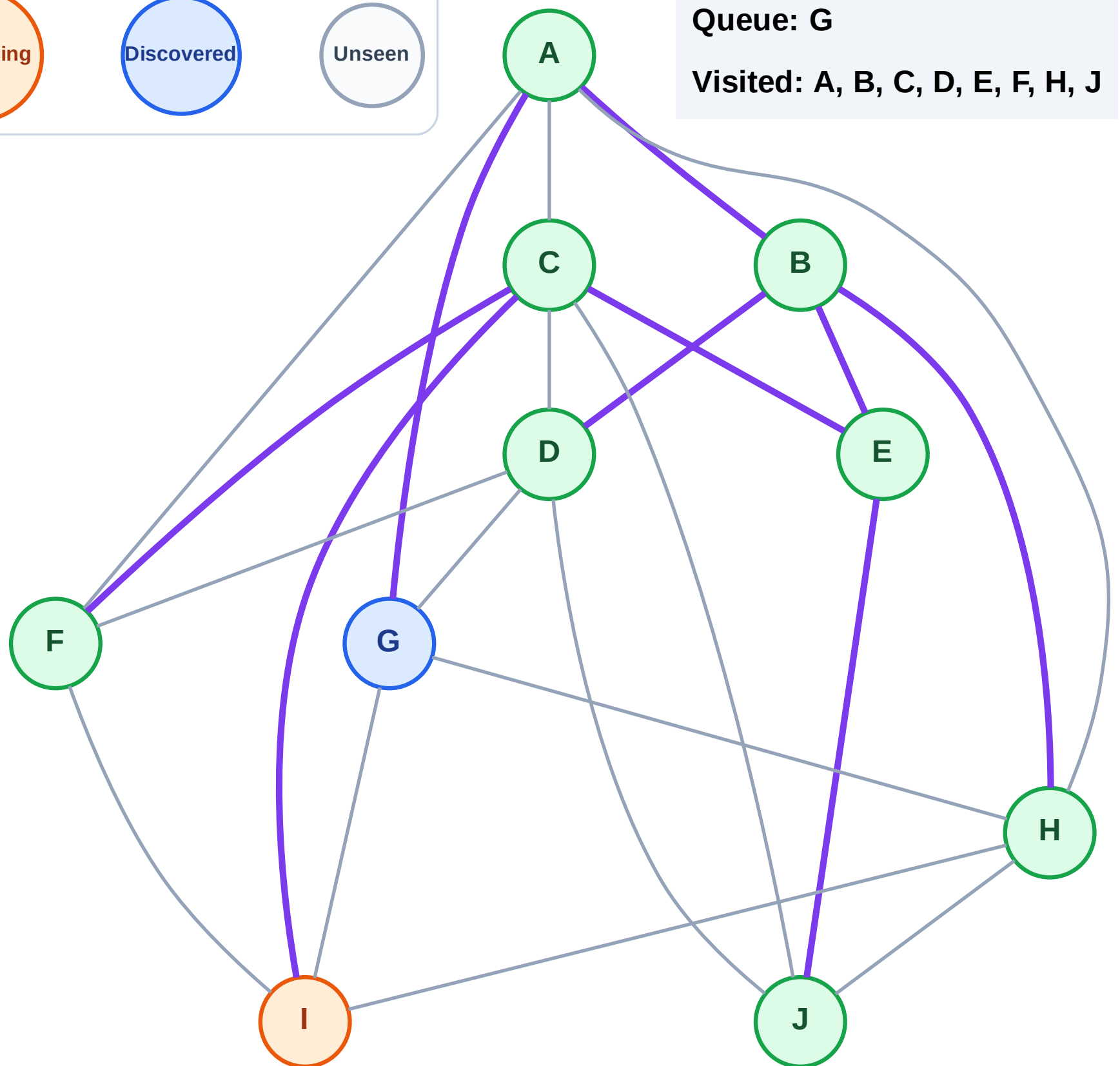
Step 18: Process node I

Legend



Queue: G

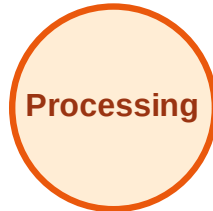
Visited: A, B, C, D, E, F, H, J



BFS Traversal

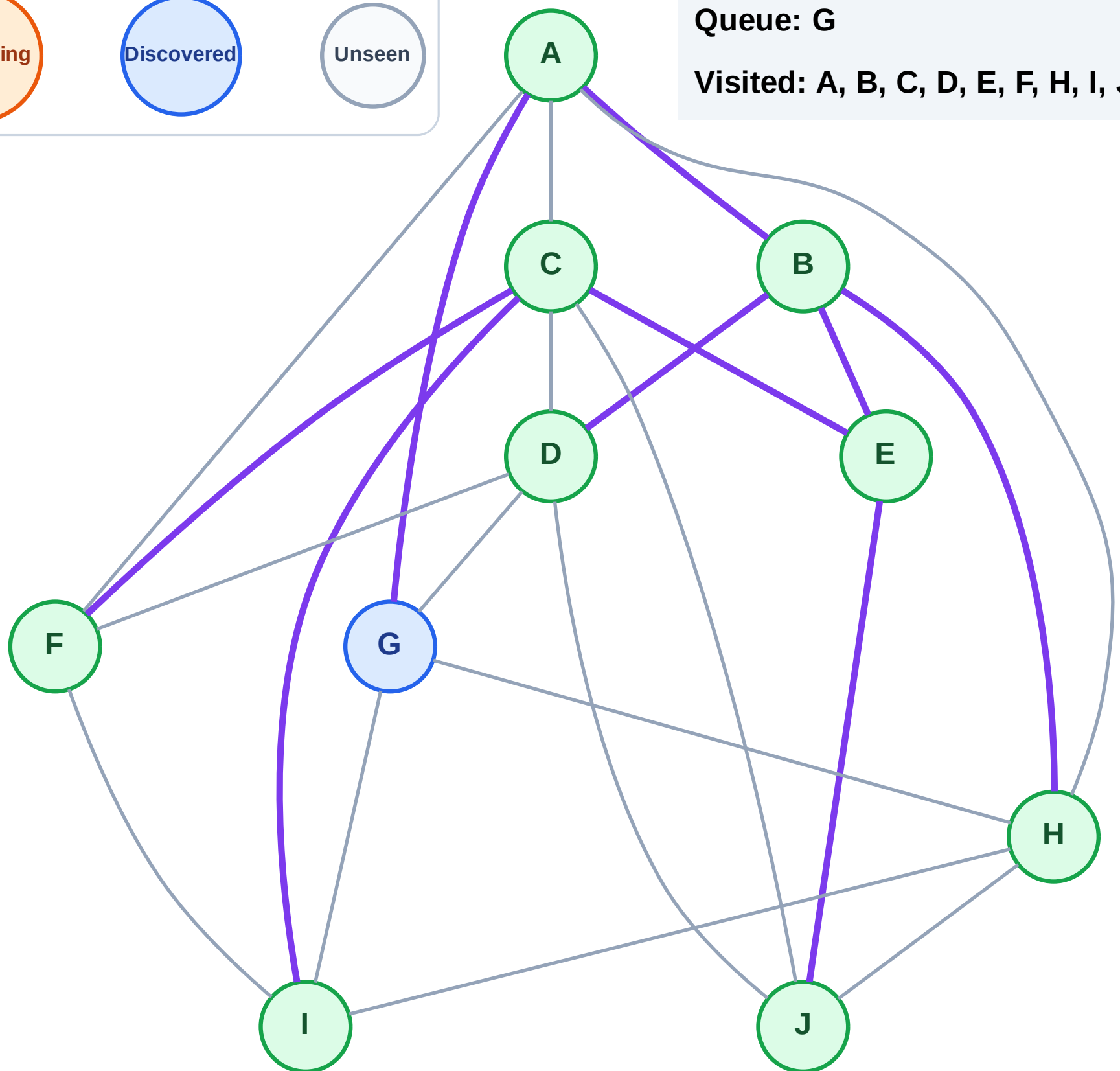
Step 19: No new neighbors from I

Legend



Queue: G

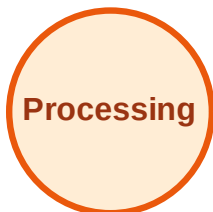
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

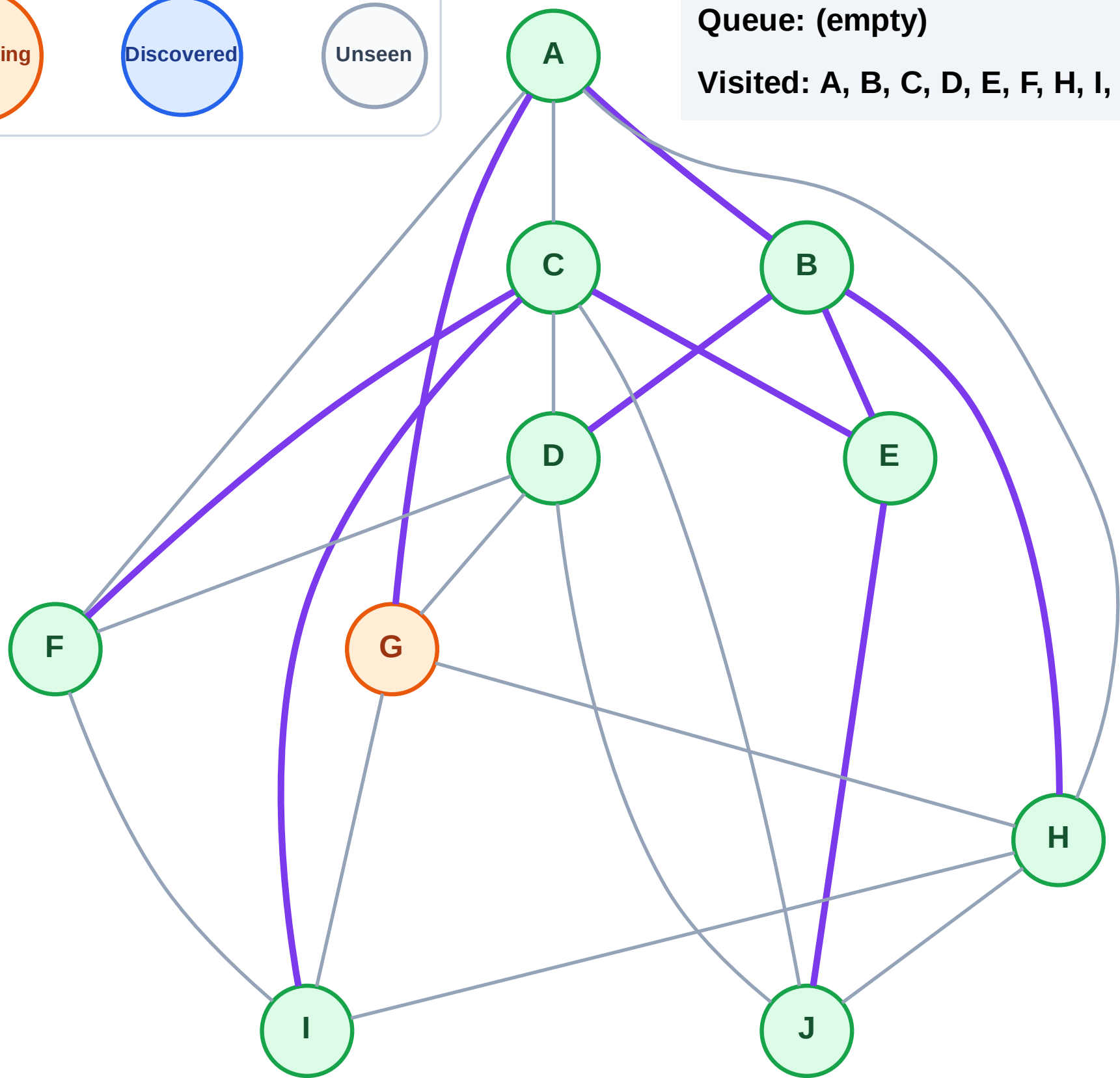
Step 20: Process node G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

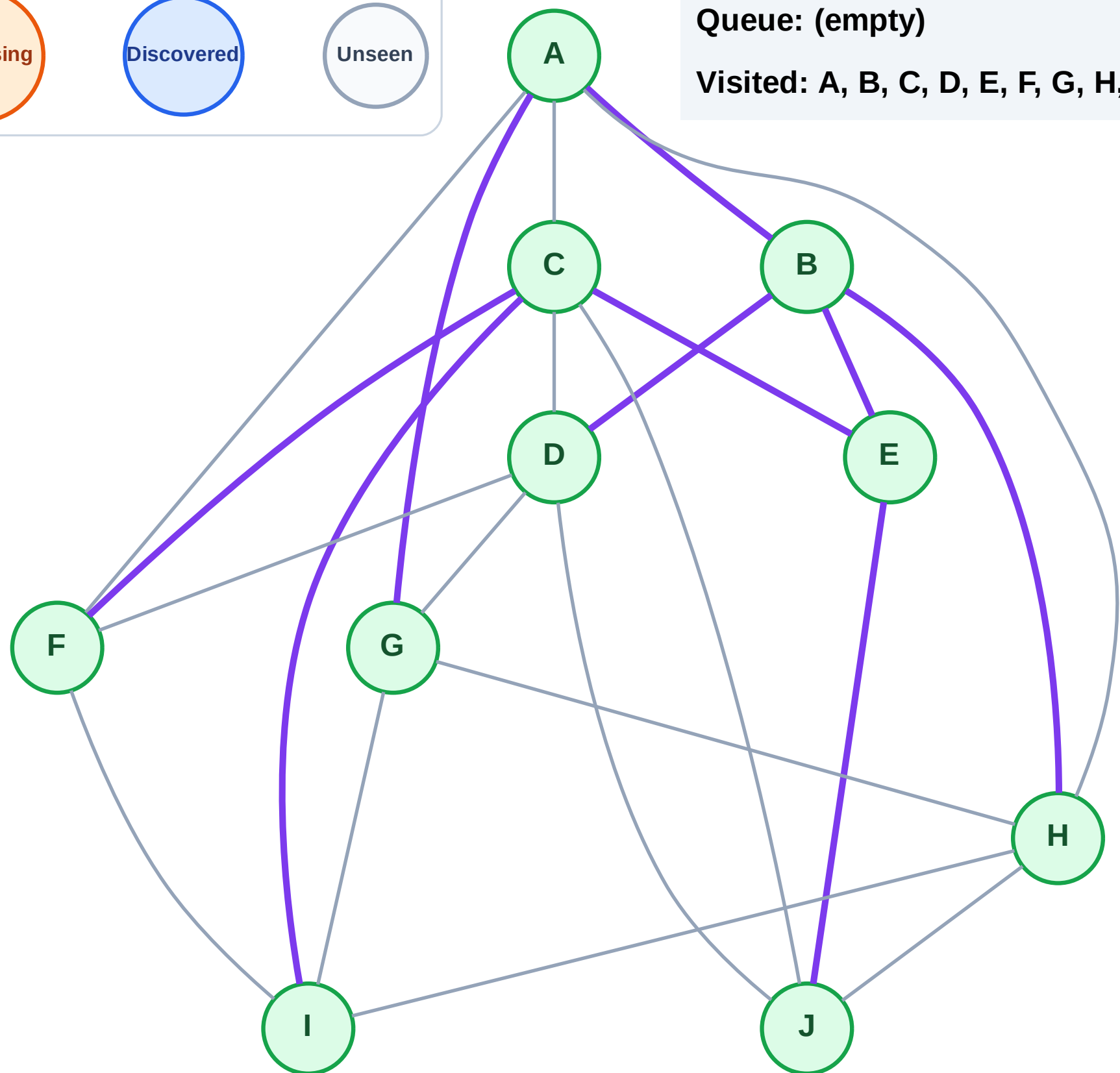
Step 21: No new neighbors from G

Legend



Queue: (empty)

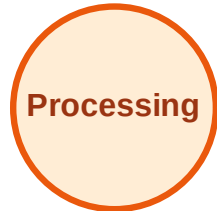
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

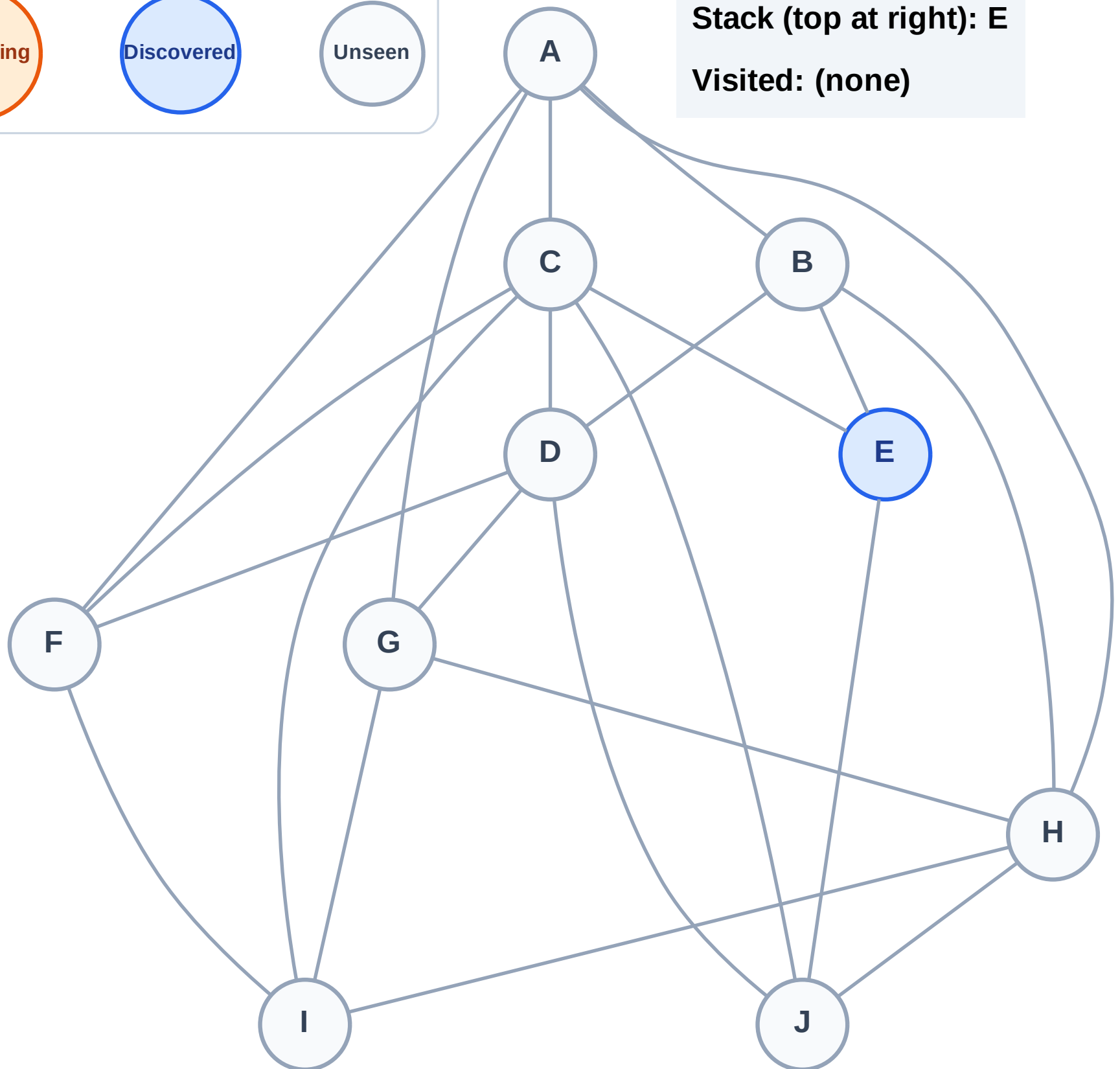
Step 1: Start at E

Legend



Stack (top at right): E

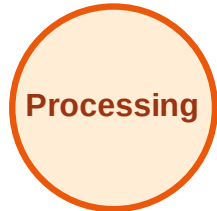
Visited: (none)



DFS Traversal

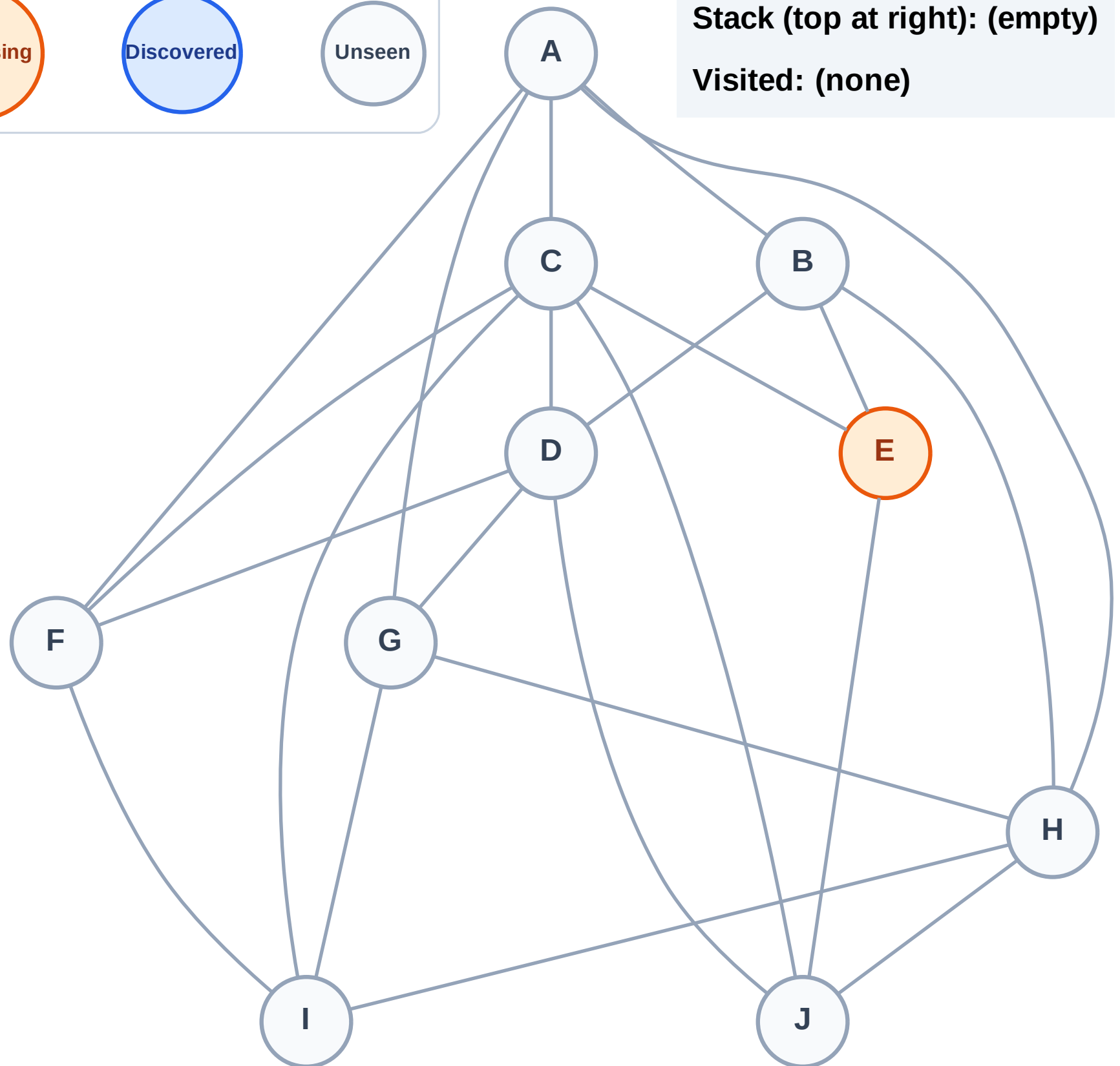
Step 2: Process node E

Legend



Stack (top at right): (empty)

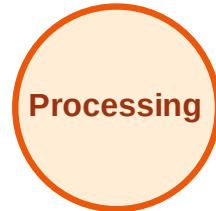
Visited: (none)



DFS Traversal

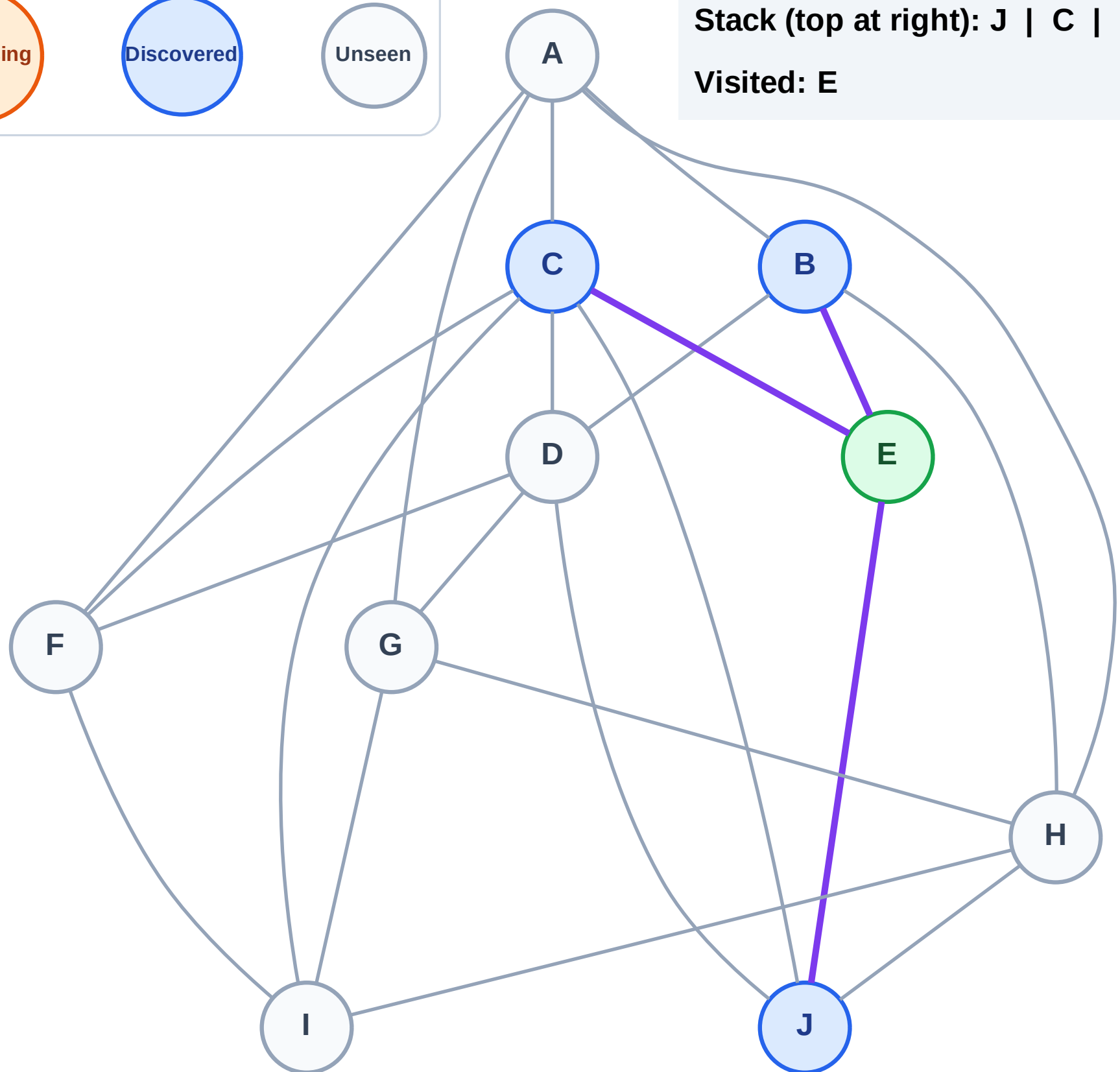
Step 3: Discover J, C, B from E

Legend



Stack (top at right): J | C | B

Visited: E



DFS Traversal

Step 4: Process node B

Legend

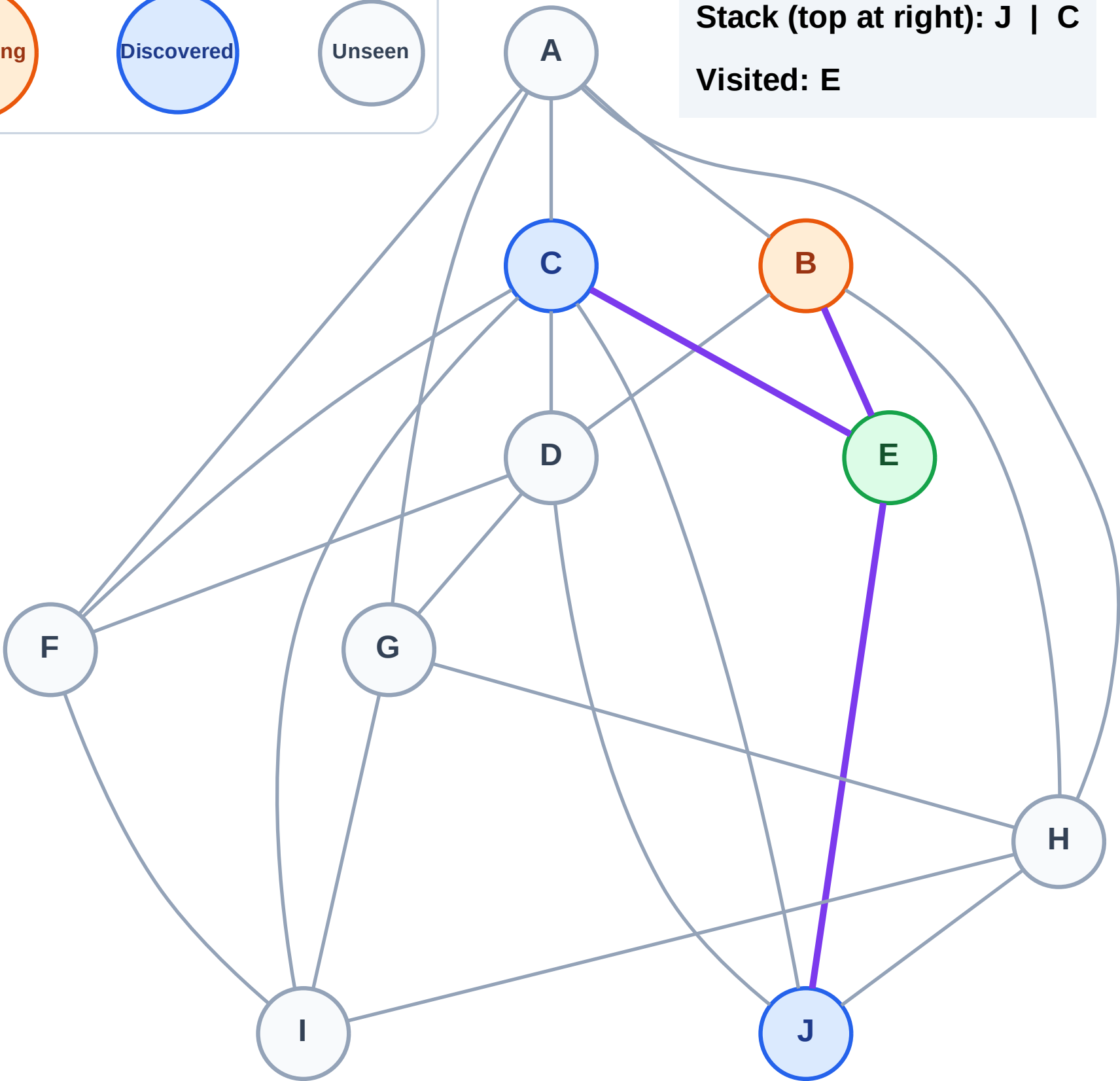
Complete

Processing

Discovered

Unseen

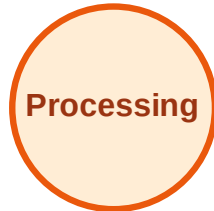
Stack (top at right): J | C
Visited: E



DFS Traversal

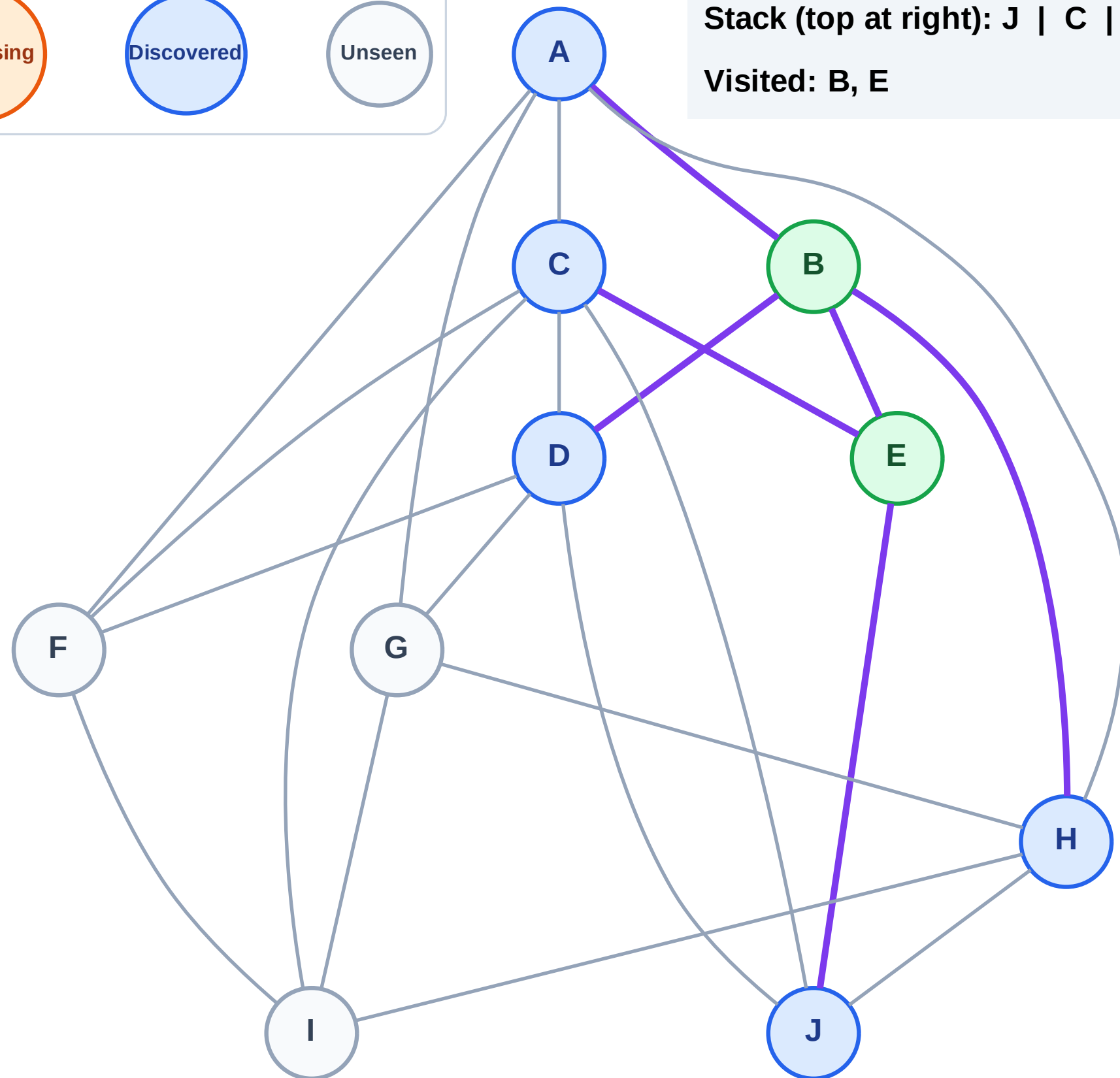
Step 5: Discover H, D, A from B

Legend



Stack (top at right): J | C | H | D | A

Visited: B, E



DFS Traversal

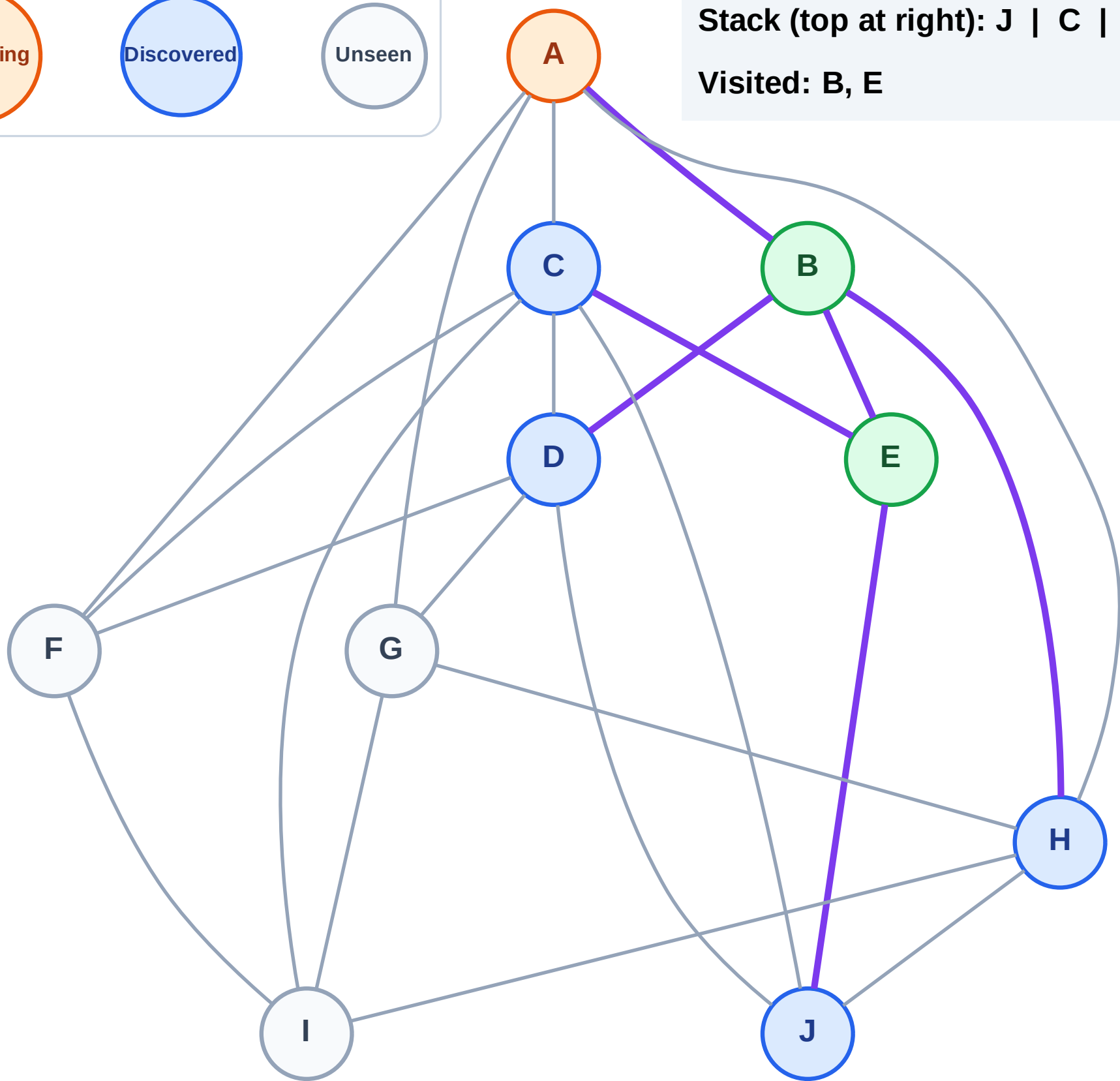
Step 6: Process node A

Legend



Stack (top at right): J | C | H | D

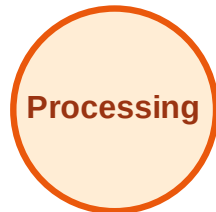
Visited: B, E



DFS Traversal

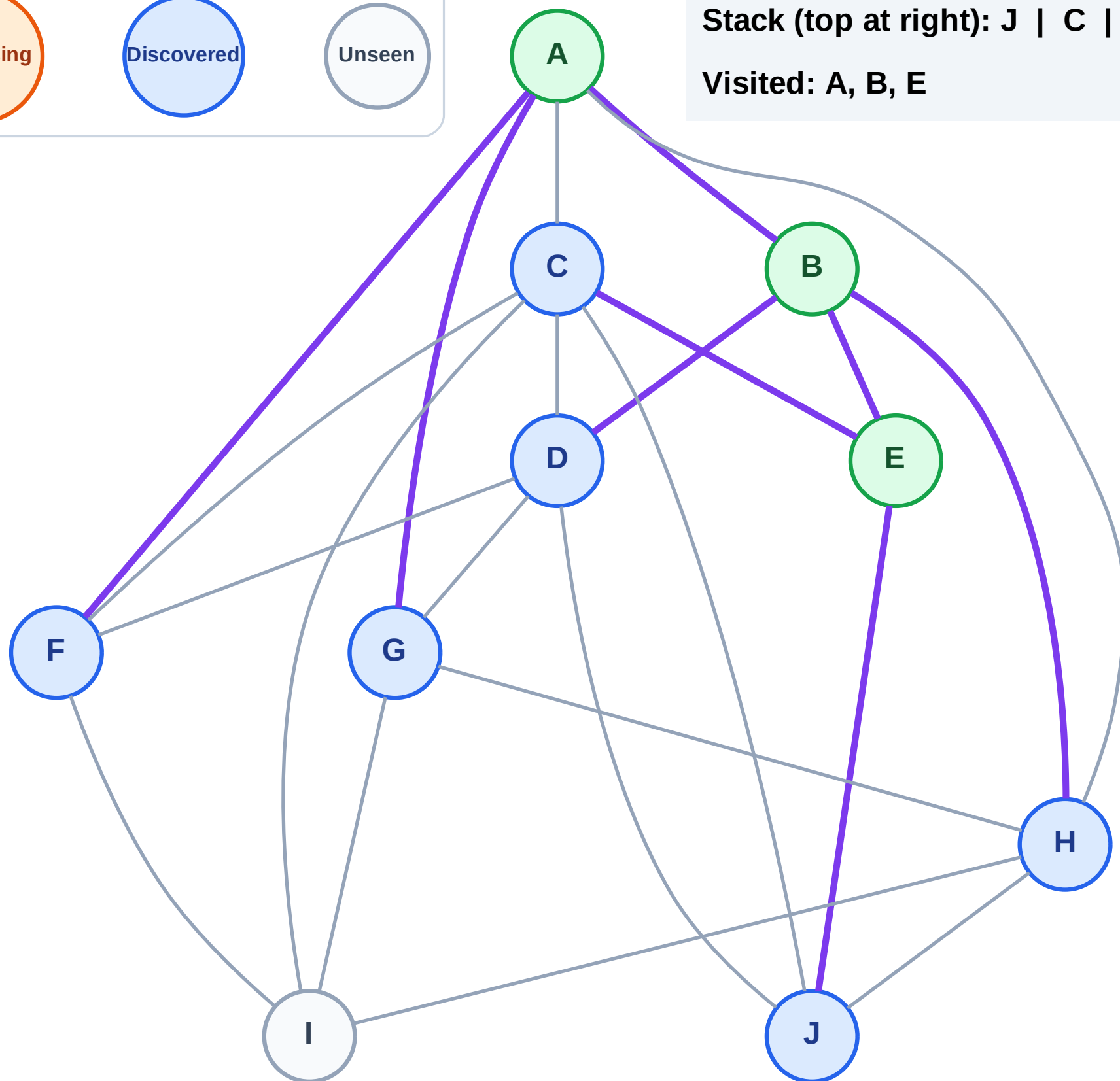
Step 7: Discover G, F from A

Legend



Stack (top at right): J | C | H | D | G | F

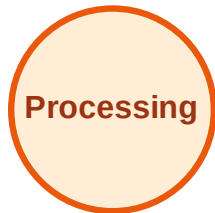
Visited: A, B, E



DFS Traversal

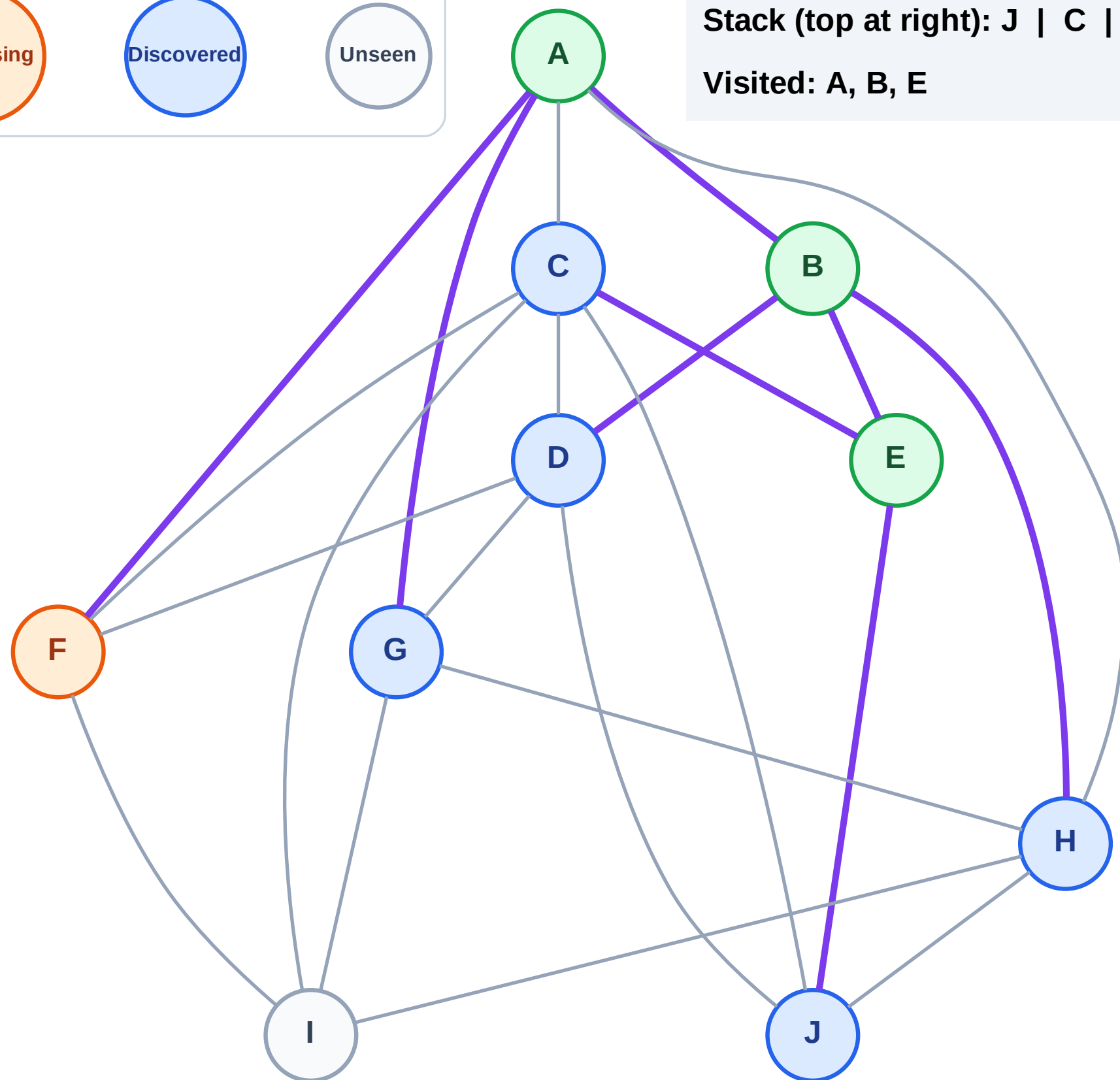
Step 8: Process node F

Legend



Stack (top at right): J | C | H | D | G

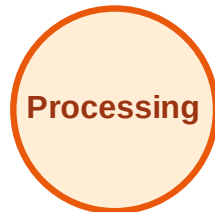
Visited: A, B, E



DFS Traversal

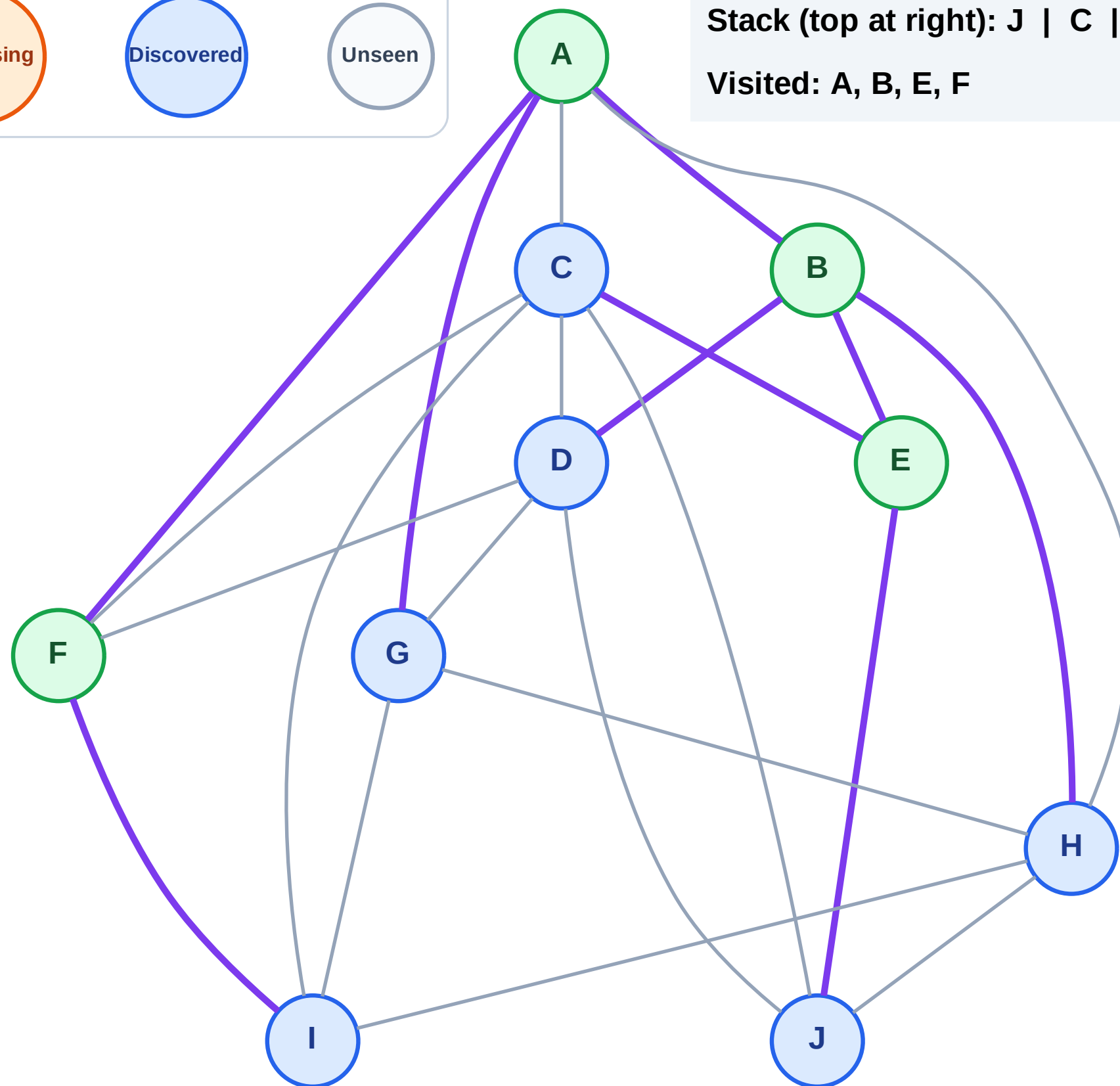
Step 9: Discover I from F

Legend



Stack (top at right): J | C | H | D | G | I

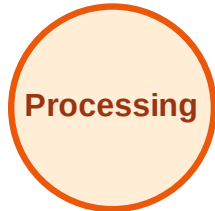
Visited: A, B, E, F



DFS Traversal

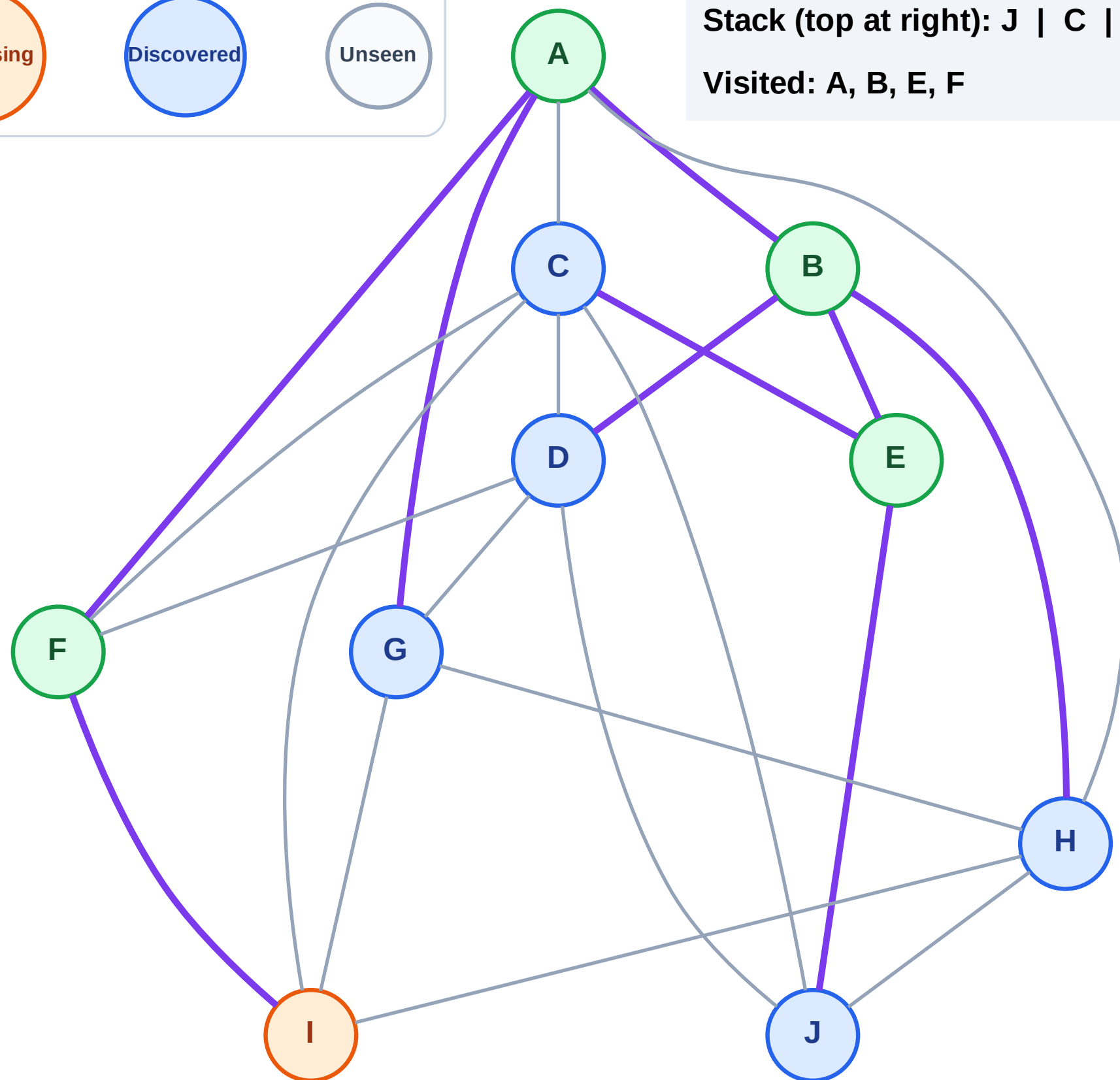
Step 10: Process node I

Legend



Stack (top at right): J | C | H | D | G

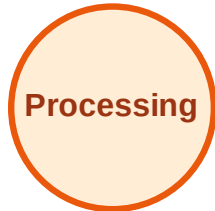
Visited: A, B, E, F



DFS Traversal

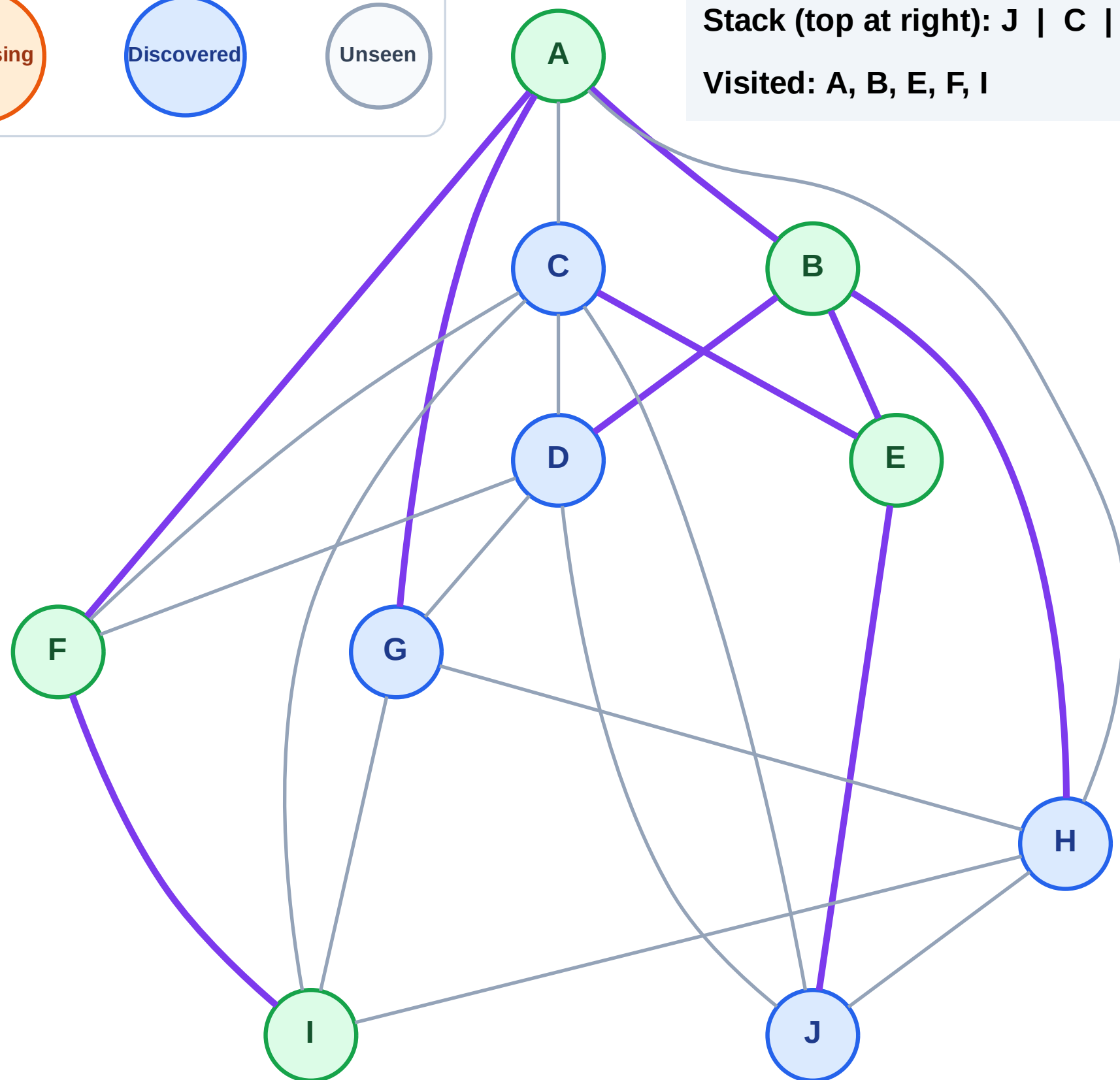
Step 11: No new neighbors from I

Legend



Stack (top at right): J | C | H | D | G

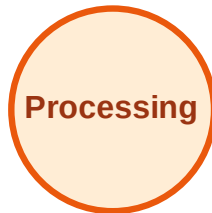
Visited: A, B, E, F, I



DFS Traversal

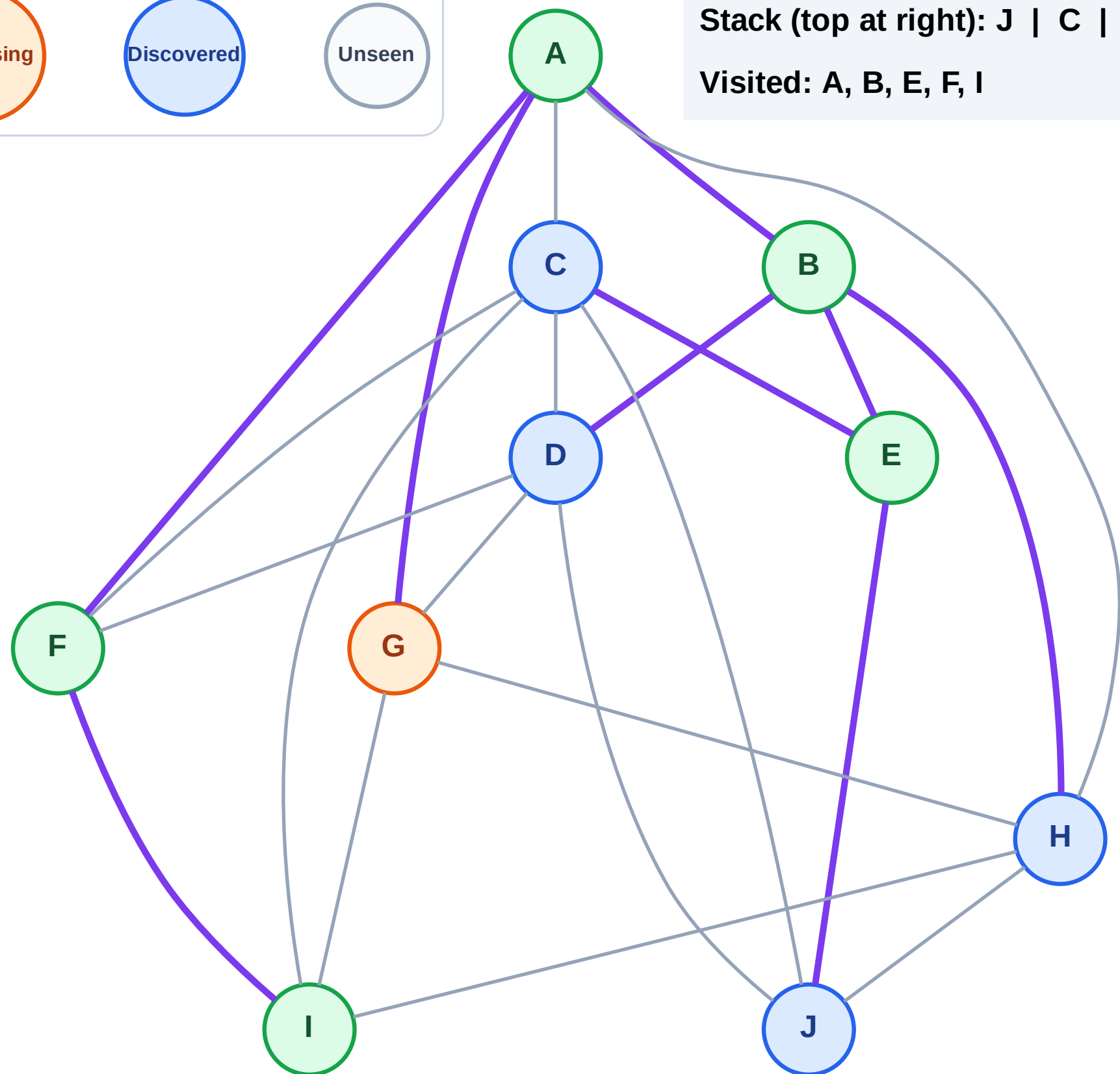
Step 12: Process node G

Legend



Stack (top at right): J | C | H | D

Visited: A, B, E, F, I



DFS Traversal

Step 13: No new neighbors from G

Legend

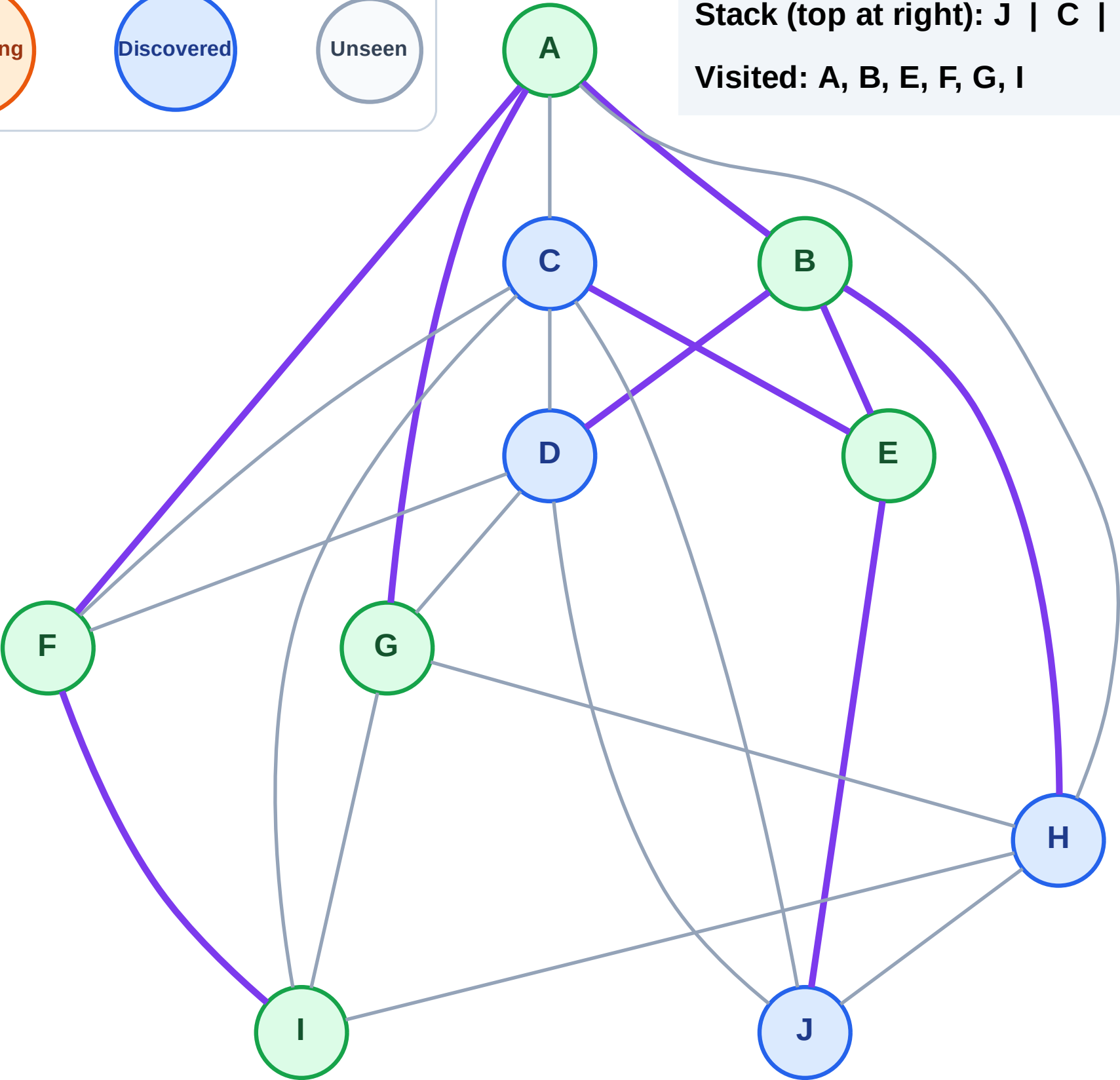
Complete

Processing

Discovered

Unseen

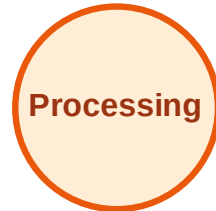
Stack (top at right): J | C | H | D
Visited: A, B, E, F, G, I



DFS Traversal

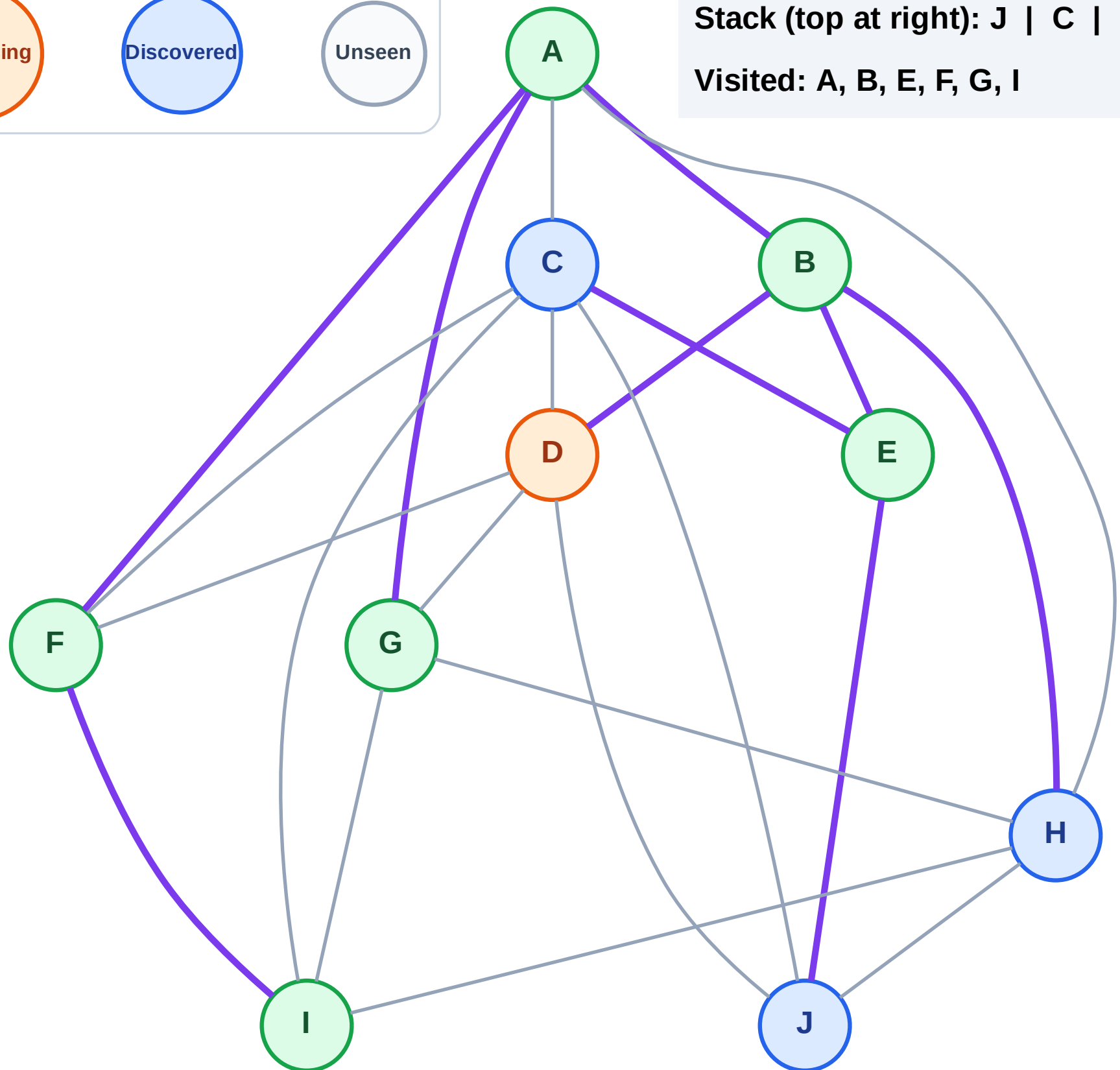
Step 14: Process node D

Legend



Stack (top at right): J | C | H

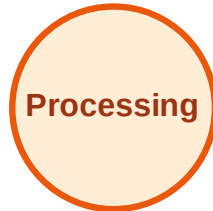
Visited: A, B, E, F, G, I



DFS Traversal

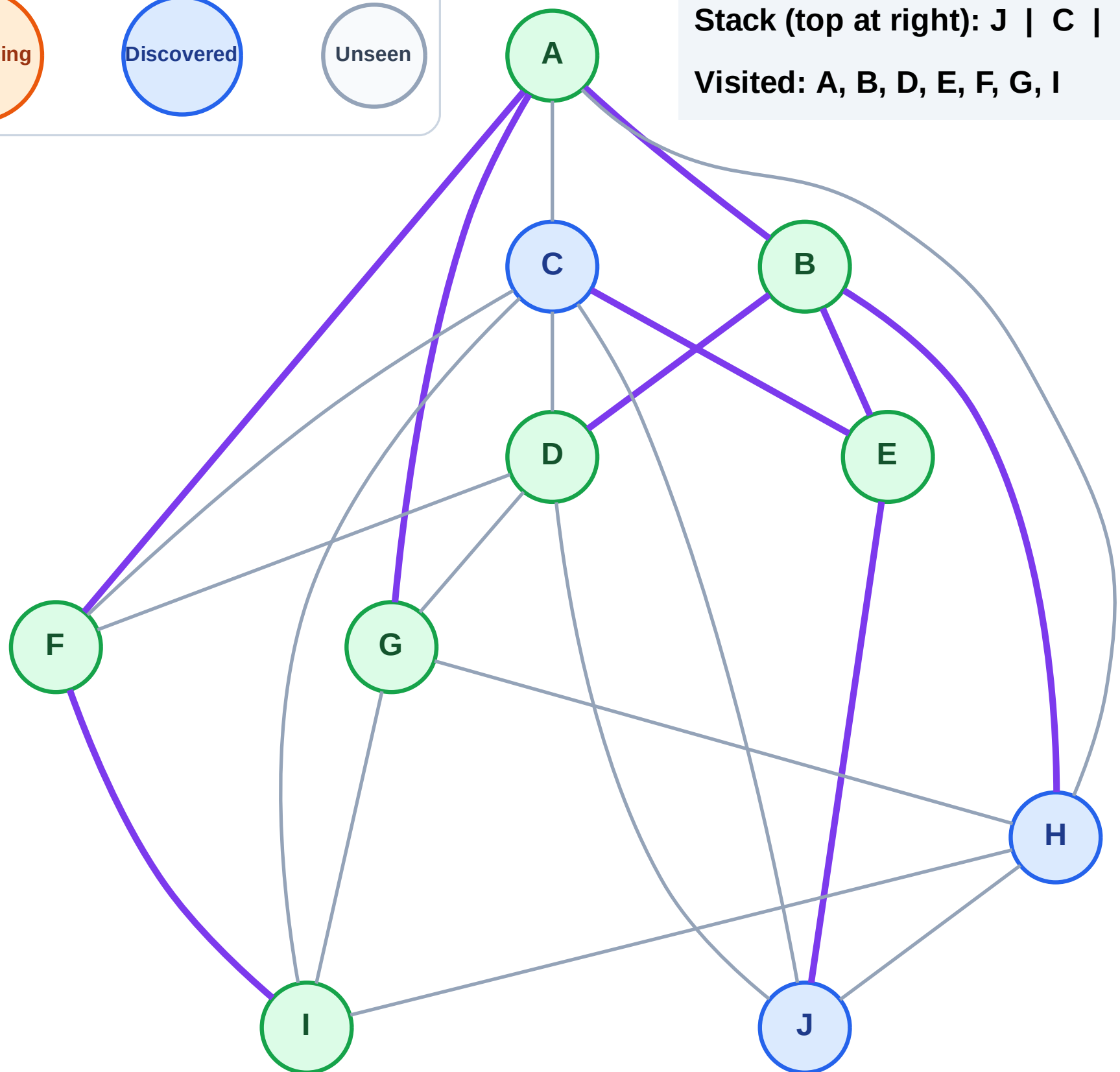
Step 15: No new neighbors from D

Legend



Stack (top at right): J | C | H

Visited: A, B, D, E, F, G, I



DFS Traversal

Step 16: Process node H

Legend

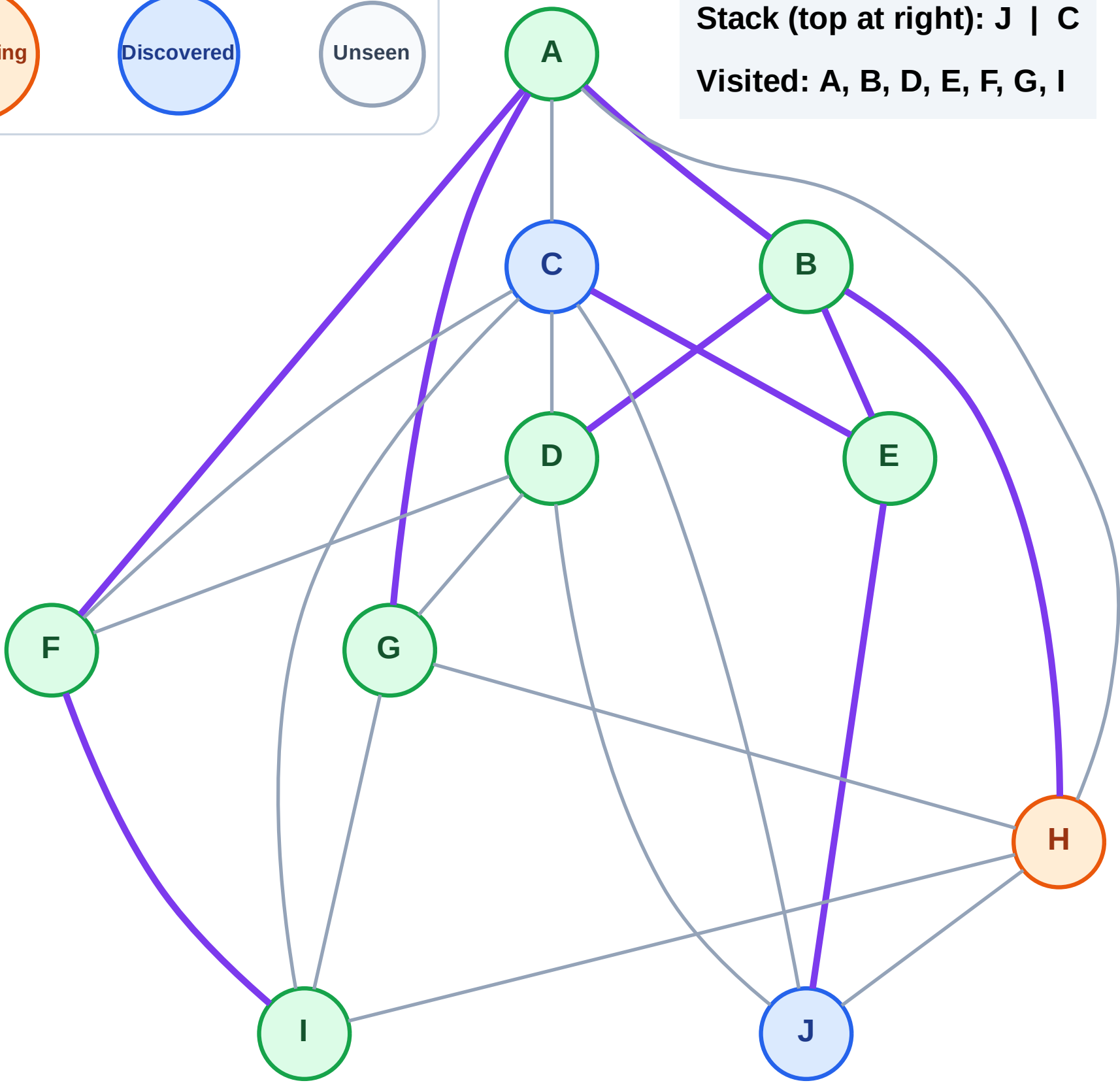
Complete

Processing

Discovered

Unseen

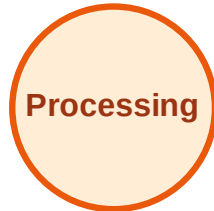
Stack (top at right): J | C
Visited: A, B, D, E, F, G, I



DFS Traversal

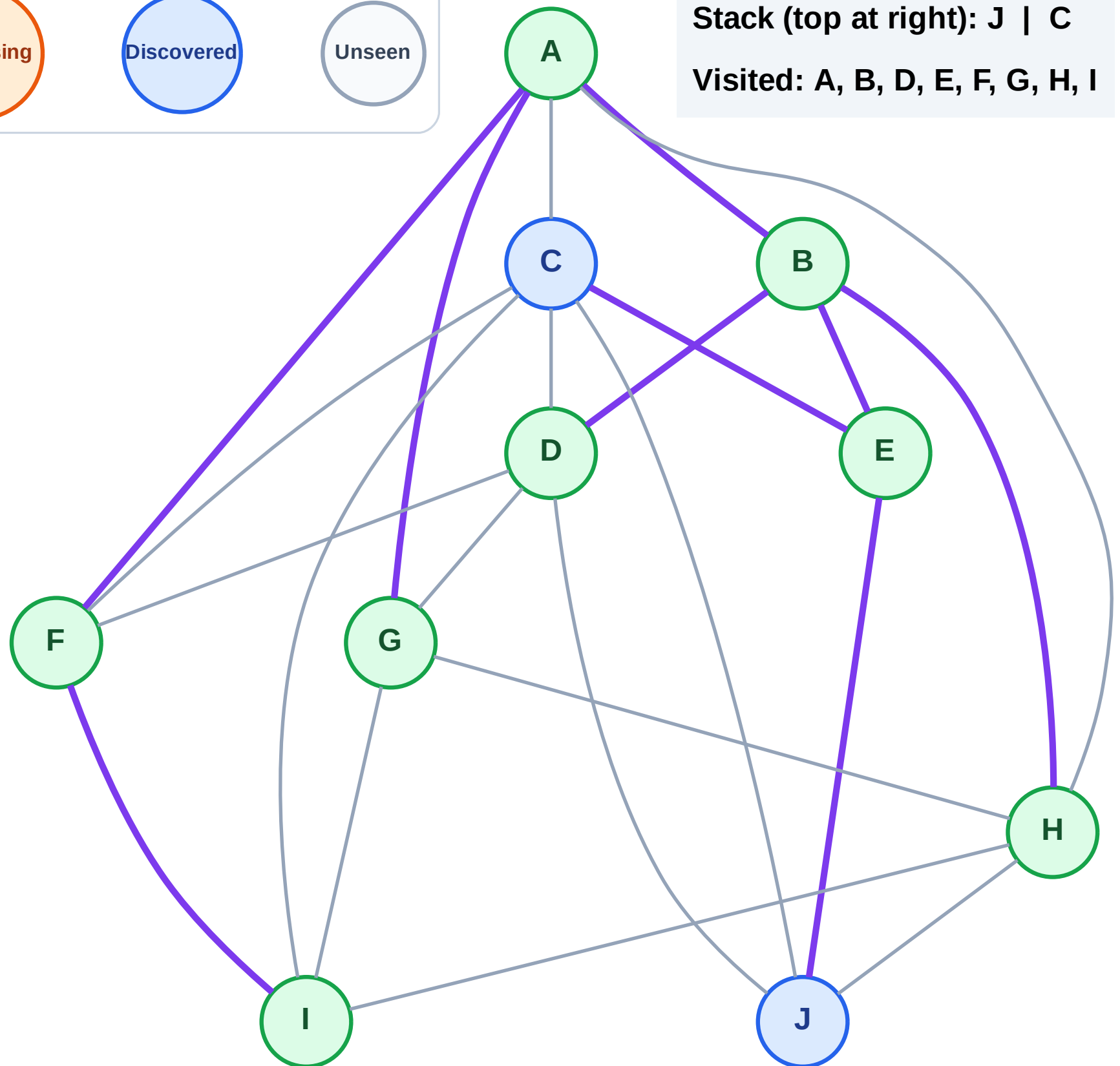
Step 17: No new neighbors from H

Legend



Stack (top at right): J | C

Visited: A, B, D, E, F, G, H, I



DFS Traversal

Step 18: Process node C

Legend

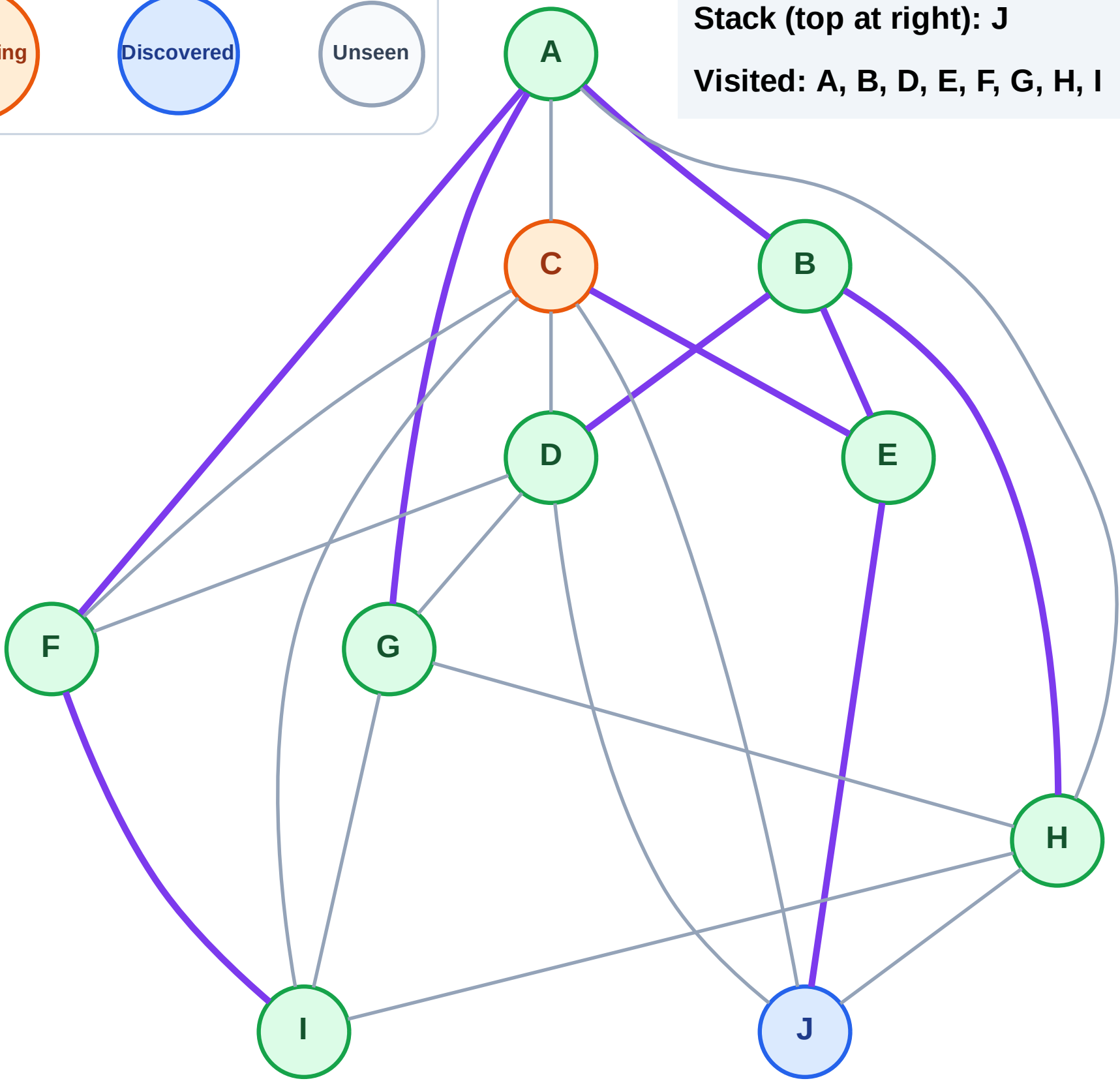
Complete

Processing

Discovered

Unseen

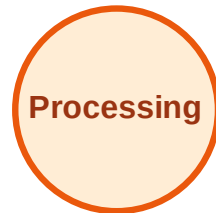
Stack (top at right): J
Visited: A, B, D, E, F, G, H, I



DFS Traversal

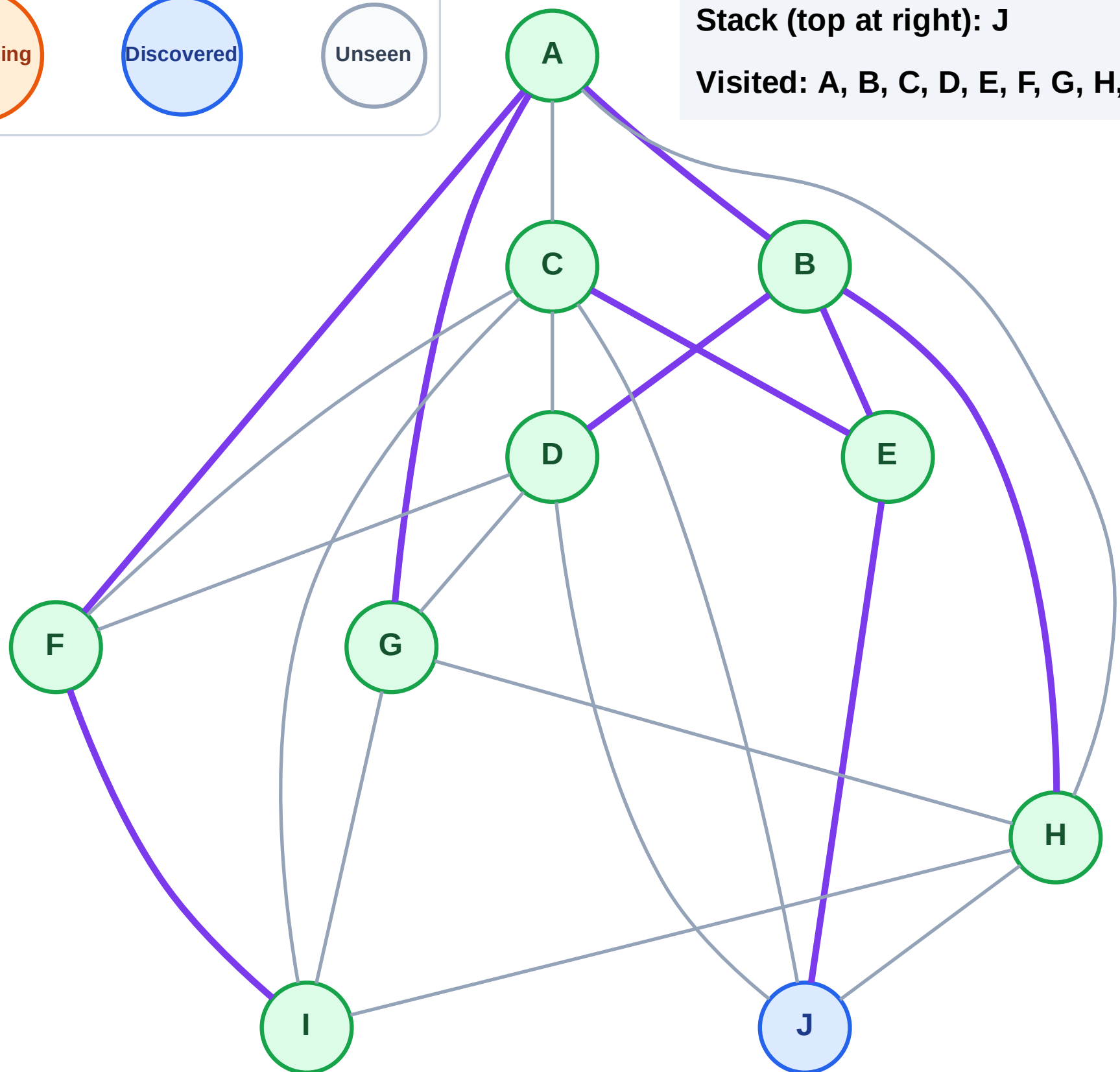
Step 19: No new neighbors from C

Legend



Stack (top at right): J

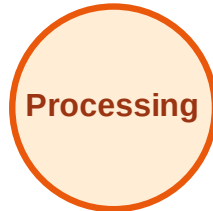
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

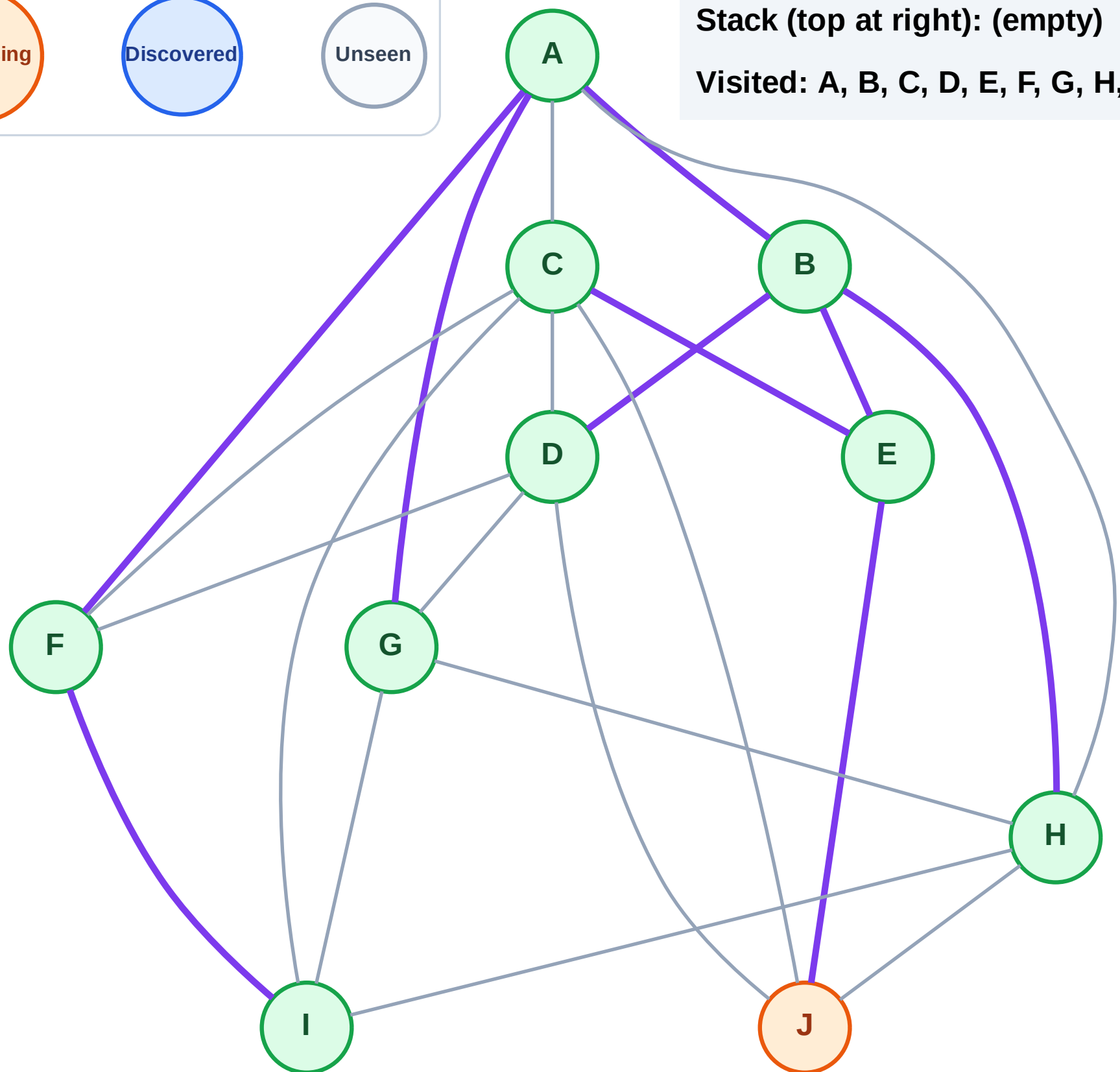
Step 20: Process node J

Legend



Stack (top at right): (empty)

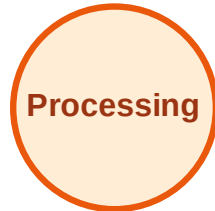
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

